

50 Stem Ideas for Girls

Inspiration, Experiments, and
Encouragements for Girls in STEM

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Author's note

Dear Reader,

This book was born from a simple observation, and a stubborn hope. Ever since my secondary school years, I have been the minority in the classroom - the Math specialization class, the STEM high school, the Computer Science, or Honors class. This doesn't happen because girls aren't curious. And definitely not because they aren't capable. But somewhere along the way, too many of them received a quiet, subtle, and damaging message: *This is not for you*.

That message is wrong. Completely, entirely, and demonstrably.

STEM - Science, Technology, Engineering, and Mathematics - is one of the most human things we do. It is the act of looking at the world around us and curiously asking, Why does it work the way it does? Could it work better? What happens if I try something differently? There is no line preventing one gender, or one type of person, from asking these questions. You were born curious. And so you ask these questions.

This book compiles 50 ideas with one rule in mind: being accessible. I started off without a lab at home or school, without a mentor who understands my struggles, without parents who work in science from whom I could seek advice. I didn't learn to code from 10. I didn't go for science camps or get lab exposure at all until high school, which was a very fortunate opportunity that I don't believe is accessible for many. But none of these should be allowed to limit you from loving the nonstop process of wondering, testing, and discovering. You have what it takes. STEM is your playground.

Just a word of caution though. STEM is definitely no easy path to take up. The challenges are real. Some concepts are very complex to grasp, or you may find yourself confused, rereading the same concept over and over until your eyes are tired. But I have loved that process as much as I hated it, always. Girls or boys (or others) alike,

choosing STEM means choosing to go through these challenges altogether. Researchers have a drawer of results that didn't work. Engineers have designs that collapse before they're brought to use. Mathematicians have proof that they had to scrap and start over. The difference between someone who succeeds in STEM and someone who gives up is rarely talent. It is almost always persistence - the ability to say, *Okay, that didn't work. Let me try again.* Frustratedly. Tiredly. But never, *I give up.*

This book doesn't require you to be good at science yet. Just your curiosity, your willingness to try something, even when unsure if it will work. The experiments here can be done at home, at school, with friends, or alone. Some take ten minutes. Some might take a weekend. Some will make a mess (Put down some newspapers first). Some will surprise you. All of them will teach you something - not just science, but about the most important STEM tool of all: your mind.

This is for you.

With much encouragement,
Vy

Why STEM matters for girls

Why STEM? Why Now? Why You?

Let's start with a number: by 2030, it is estimated that more than 75% of the fastest-growing careers in the world will require skills in science, mathematics, technology, or engineering. These are the jobs that will build the hospitals of the future, design the clean energy systems that power our cities, write the software that connects billions of people, and develop the medicines that will treat diseases we haven't even named yet.

These are good jobs — meaningful, creative, and well-paid. And right now, women hold fewer than 30% of them.

That is not because women and girls are less capable. Decades of research have shown that girls perform equally — and often better — than boys in math and science during their school years. The gap is not in ability. The gap is in encouragement, representation, and opportunity.

This matters beyond careers, too. STEM teaches ways of thinking that are useful everywhere:

Critical thinking. STEM teaches you to ask, *Is this claim actually supported by evidence?* This skill protects you from misinformation, helps you make better decisions, and makes you a clearer communicator.

Problem-solving. In STEM, you learn to break a big, impossible-seeming problem into small, manageable steps. This is not just useful in a laboratory — it's useful in every challenge life throws at you.

Creativity. The best scientists are deeply creative people. Designing an experiment, figuring out why something isn't working, or imagining a new way to use existing

materials all require imagination. STEM and creativity are not opposites — they are partners.

Resilience. Perhaps most importantly: STEM teaches you how to fail well. Every hypothesis that gets disproven, every bridge model that collapses, every line of code that throws an error — these are not failures. They are data. They tell you what to try next.

When more girls enter STEM, the whole world benefits. Research consistently shows that diverse teams produce better science. When women are at the table, different questions get asked — questions that lead to discoveries and solutions that might otherwise be missed. The COVID-19 vaccine was developed at record speed partly because of the contributions of Dr. Katalin Karikó, a woman who spent decades working on mRNA technology while many doubted her. The first images of a black hole were made possible by the algorithms of Dr. Katie Bouman, who developed the imaging methods as a graduate student. When girls step into STEM, history gets made.

STEM Is Not "Just for Boys"

We need to say this clearly, because many girls have received the opposite message from school, from media, from toy stores, from the internet, and sometimes — unintentionally — from the people around them:

STEM has never been only for boys.

The stereotype of the lone male scientist in a lab coat is not accurate history. It is a story constructed over time by excluding women from credit, from recognition, and from the institutions where science was practiced. When we look more carefully at the history of science, we find women everywhere — often working behind the scenes, often uncredited, and often doing work that was central to the most important discoveries of their time.

Rosalind Franklin's X-ray crystallography images were used — without her full knowledge or credit — to help Watson and Crick identify the double helix structure of DNA. Lise Meitner co-discovered nuclear fission and was passed over for the Nobel Prize that went to her male collaborator. Cecilia Payne-Gaposchkin discovered that stars are mostly made of hydrogen and helium — a finding that revolutionized astronomy — and was initially told by her (male) supervisor that she must be wrong.

These women were brilliant. They were rigorous. They were right. And the history of science is full of them.

It is also worth noting that in many non-Western cultures and historical periods, women held prominent places in scientific and medical knowledge. This is a global history, and it belongs to all of us.

Today, women like Jennifer Doudna (co-inventor of CRISPR gene editing), Fabiola Gianotti (Director General of CERN), and Gitanjali Rao (young inventor and Time's first Kid of the Year) are rewriting what a scientist looks like.

You can be a part of its future too!

Some practical things to remember when you encounter the stereotype:

- If someone says, *girls are bad at math*, remember: math ability is not biological. It is built through practice, encouragement, and good teaching.
- If you look around a classroom and you're the only girl interested in a STEM topic, that is a problem with the classroom — not with you.
- If you don't see women who look like you in STEM, that is an invitation — not a sign to leave.

How to Use This Book

This book is organized into five parts, each focused on a different area of STEM. You do not need to read it from start to finish. You do not need to do every experiment. You do not need to complete any of them perfectly.

Here is how we suggest using it:

- **Browse first.** Flip through the book and see what catches your attention. You might find yourself drawn to the engineering challenges in Part 3, or to the environmental projects in Part 5, or to the coding ideas in Part 4. Start where your curiosity pulls you.
- **Choose one thing.** Pick one experiment or project to try. Just one. Don't overwhelm yourself. Read through it, gather your materials, and try it. See what happens.
- **Write things down.** Get a notebook — it doesn't need to be anything fancy — and use it as a science journal. Write down what you expected to happen, what actually happened, and any questions the experiment raised. Scientists call this keeping a lab notebook. It is one of the most important habits in science.
- **Don't skip the failures.** If something doesn't work, write that down too. Try to figure out why. Could your materials have been different? Did you miss a step? Was your hypothesis wrong? Investigating a failed experiment is often more educational than a successful one.
- **Invite someone.** Science is more fun with company. Try the experiments with a friend, a sibling, a parent, or a teacher. Explain what you're doing. Teaching something is one of the best ways to understand it more deeply.
- **Come back.** You don't have to do everything at once. This book is meant to be picked up again and again, over weeks and months and years. Some of the ideas here might seem too hard right now and feel just right in six months.

There is no grade at the end of this book. There is no test, no deadline, no right or wrong way to use it. There is only curiosity, and what happens when you follow it.

Let's begin.

Women Who Changed the World through STEM

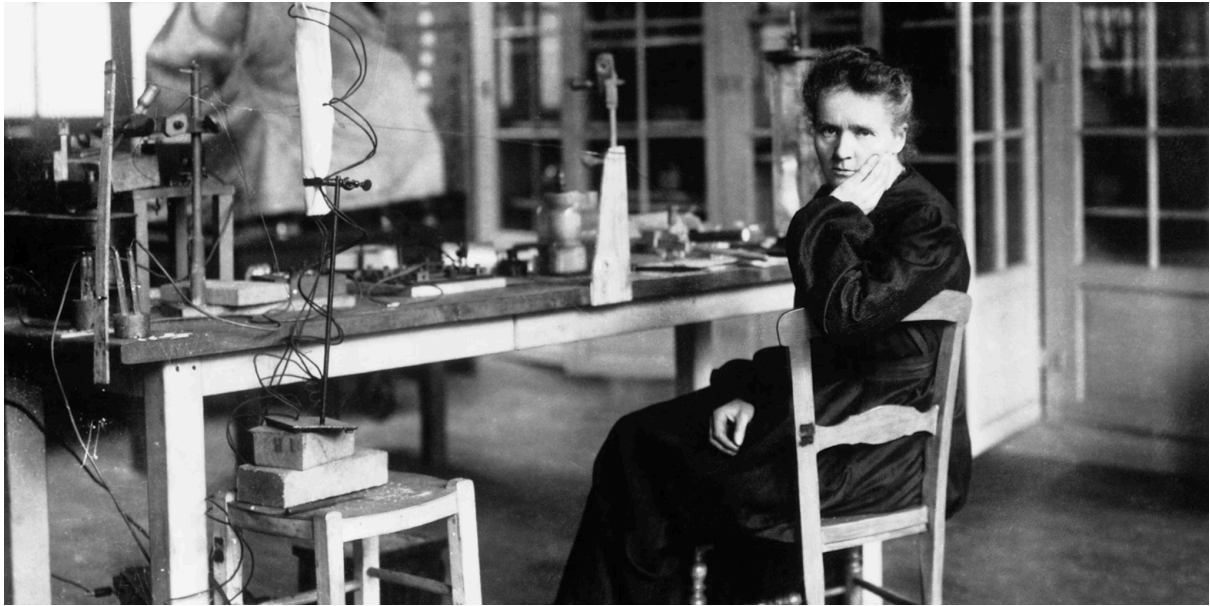
Six extraordinary women. Six lives of curiosity, courage, and discovery.

1. Marie Curie (1867–1934) |Physicist and Chemist | Poland and France

Marie Curie is perhaps the most famous woman in the history of science — and for very good reason. Born Maria Skłodowska in Warsaw, Poland, she grew up in a country under foreign rule where women were barred from attending university. Rather than accept this, she made a deal with her sister: each would support the other's education in turns. Marie worked as a governess for years, sending money to her sister in Paris, before finally making the journey herself.

In Paris, she studied physics and mathematics at the Sorbonne, often in poverty, sometimes so cold in her rented room that the water in her washbasin froze overnight. She graduated first in her class. She met and married Pierre Curie, and together they began researching the mysterious rays emitted by uranium — a phenomenon she herself named *radioactivity*.

Her discoveries were staggering. She identified two new elements — polonium (named for her home country) and radium — and proved that radioactivity was an atomic property, not a chemical reaction. She won the Nobel Prize in Physics in 1903 (the first woman ever to receive one, shared with Pierre and Henri Becquerel), and the Nobel Prize in Chemistry in 1911 — becoming the first person in history, male or female, to win Nobel Prizes in two different sciences.



Hulton-Deutsch Collection/CORBIS/Corbis via Getty Images

What makes Curie an inspiration is not just her intelligence, but her persistence. She worked in a shed. She was denied positions because of her gender. Her discoveries were sometimes met with skepticism. She kept going anyway. And when World War I broke out, she drove mobile X-ray units to the front lines, saving soldiers' lives.

What she shows us: Obstacles are not verdicts. They are problems to be solved.

2. Ada Lovelace (1815–1852) | Mathematician | England



Ada Lovelace lived in the 19th century, more than a hundred years before computers existed — and yet she is widely regarded as the world's first computer programmer.

Ada was the daughter of the poet Lord Byron, though she never knew him. Her mother, determined that Ada would not inherit her father's wild temperament, gave her an unusual-for-girls education heavily focused on mathematics. Ada showed an extraordinary gift, and at seventeen she met Charles Babbage, a mathematician who was designing a theoretical calculating machine he called the Analytical Engine.

Ada understood Babbage's machine — possibly better than Babbage himself did. When she translated an Italian article about the Analytical Engine into English, she added her own notes, which were three times longer than the original article. Those notes included what is now recognized as the first algorithm designed to be processed by a machine — making her the first person in history to describe what we now call software.

She also had a breathtaking vision of what computing could eventually become. She wrote that the Analytical Engine might one day compose music, produce graphics, and be used for scientific purposes — predictions that proved remarkably accurate, though it would take another century for anyone to build what she had imagined.

Ada died at just 36, never seeing her vision realized. But the programming language ADA, developed by the U.S. Department of Defense, was named in her honor. And every time you use software — every app, every website, every game — you are living in the world she first imagined.

What she shows us: Imagination and mathematics are not separate. The greatest thinking happens when they work together.

3. Katherine Johnson (1918–2020) | Mathematician | United States

Katherine Johnson calculated the flight trajectories for some of the most important moments in space exploration history — including Alan Shepard's first American

trip to space, John Glenn's orbit of Earth, and the Apollo 11 mission that landed humans on the moon.

She did this at a time when Black women in America were being told, by law and by custom, that many spaces were not for them. She worked at NASA (then called NACA) as a "human computer" — a term used for the teams of women, many of them Black, who performed the complex mathematical calculations that electronic computers were not yet trusted to do reliably.



John Glenn, one of the most celebrated astronauts in NASA history, famously refused to fly his orbital mission until Katherine Johnson personally verified the numbers that the electronic computer had produced. He trusted her mathematical precision over the machine. That trust was well placed. Glenn's mission succeeded.

Katherine Johnson worked at NASA for 35 years, authoring or co-authoring 26 scientific papers. In 2015, President Barack Obama awarded her the Presidential Medal of Freedom. She was portrayed in the 2016 film **Hidden Figures**, which finally brought her story to a wide audience — though she had been performing this extraordinary work, largely unsung, for decades.

She once said: *"I felt most proud of being accurate."* In a field where being off by a fraction of a degree could mean the difference between a spacecraft reaching the moon or disappearing into space, accuracy was everything. And she had it.

What she shows us: Precision is a superpower. And doing your job excellently, even when the world isn't watching, matters.

4. Rosalind Franklin (1920–1958) | Chemist and X-ray Crystallographer | England



Credit: Photo Researchers/Science History Images/Alamy

Rosalind Franklin is one of the most important — and most unjustly overlooked — figures in the history of biology. Using a technique called X-ray crystallography, she produced an image known as Photo 51: an X-ray diffraction photograph of DNA that is widely regarded as one of the most important images in the history of science.

That photograph revealed the helical structure of DNA — the double helix shape that carries the genetic information of all living things. When Watson and Crick announced their model of DNA in 1953, it was based significantly on Photo 51 and data from Franklin's work — data that was shared with them without her knowledge or permission. When Watson and Crick, along with Franklin's supervisor Maurice Wilkins, won the Nobel Prize in 1962, Franklin had already died of ovarian cancer at age 37. Nobel Prizes are not awarded posthumously, and her contribution was barely acknowledged in the speeches.



Maurice Wilkins (left), James Watson and Francis Crick at the ceremony for the 1962 Nobel Prize in Physiology or Medicine. Credit: King's College London Archives: K/PP178/15/3/1

Franklin's story is one of the starkest examples of how women's scientific contributions have been erased and uncredited throughout history. But her work stands apart from how it was treated. She was a meticulous, brilliant experimentalist whose techniques advanced the understanding of molecular structure far beyond DNA — she also made major contributions to the study of viruses, including tobacco mosaic virus and polio.

What she shows us: The quality of your work matters, independently of whether you get credit for it. History has a way of finding the truth — and Rosalind Franklin's name is now permanently part of it.

5. Mae Jemison (1956–present) | Engineer, Physician, and Astronaut | United States

On September 12, 1992, Mae Jemison became the first Black woman in history to travel to space. She flew aboard the Space Shuttle Endeavour as a mission specialist, conducting experiments in bone cell research and human adaptation to weightlessness. She holds the distinction of having worn two professional hats that most people would consider extraordinary on their own: she is both a medical doctor and an engineer.



Mae Jemison aboard the space shuttle *Endeavour*.

Jemison grew up in Chicago and was interested in science from early childhood. She earned a chemical engineering degree from Stanford University and a medical degree from Cornell, served in the Peace Corps as a medical officer in Sierra Leone and

Liberia, and then applied to NASA's astronaut program. She was selected from more than 2,000 applicants.

After leaving NASA, Jemison founded several companies focused on education and technology, including a science camp for students that she runs to this day. She also appeared in an episode of *Star Trek: The Next Generation* — a detail that reveals something important about her: she is, in the best sense, a dreamer. She believes in imagining the future we want to build, and then building it.

She has said: *"Never be limited by other people's limited imaginations."* This is not a platitude from her — it is literally how she has lived her life.

What she shows us: You can be many things at once. The categories that others put you in are not the edges of what you can become.

6. Tu Youyou (1930–present) | Pharmaceutical Chemist | China



In the 1960s and 1970s, malaria — a parasitic disease spread by mosquitoes — was killing hundreds of thousands of people every year, and the parasites causing it were becoming resistant to existing drugs. The Chinese government launched a secret

research initiative called Project 523 to find a new treatment. Tu Youyou was the lead scientist.

Working with a team and drawing on centuries of traditional Chinese medical texts, Tu screened hundreds of herbal compounds. She found a promising substance called artemisinin in sweet wormwood — a plant that had been used in Chinese medicine for over 1,500 years. But extracting it was difficult; early results were inconsistent. Tu returned to the ancient texts and found a 4th-century account suggesting that the plant should be prepared with cold water, not heat. She adapted her extraction method — and it worked.



Tu Youyou with one of her mentors, pharmacologist Lou Zhicen, in the 1950s. Lou Zhicen trained her to identify medicinal plants based on their botanical descriptions. Photo: Public domain

Artemisinin is now the foundation of the most effective antimalarial treatments in the world. It has saved millions of lives, particularly in sub-Saharan Africa. In 2015, Tu Youyou became the first Chinese woman to win the Nobel Prize in Physiology or Medicine — at the age of 84, recognition that had taken decades to arrive.

Her work is a beautiful example of how traditional knowledge and modern science can work together: the ancient wisdom pointed the way; the modern method confirmed, refined, and scaled it.

What she shows us: Knowledge from unexpected places — including the past — can lead to the most important discoveries. Stay curious about everything.

What We Learn from Women in STEM

The six women you just met came from different countries, different centuries, and different fields of science. They faced different obstacles — poverty, racism, sexism, illness, erasure. And yet, looking across their lives, certain qualities appear again and again. These are the qualities that carried them through.

1. Curiosity — The Starting Engine

Every one of these women was, at her core, deeply curious. Marie Curie was fascinated by the invisible rays uranium produced and couldn't stop asking what they were. Ada Lovelace met a calculating machine and immediately began imagining what else it could do. Tu Youyou read a 1,500-year-old text and asked whether an ancient remedy might hold a modern answer.

Curiosity is not a character trait you either have or don't have. It is a habit you can practice. When you encounter something you don't understand, you have two choices: let it pass, or pause and ask **why**. The scientists we admire chose to pause.

In your STEM journey, curiosity is your most important tool. More important than any piece of equipment, any textbook, or any grade. If you leave every experiment with at least one new question — something you noticed, something that confused you, something you want to understand better — then you are doing science.

2. Persistence — Staying When It Gets Hard

None of the breakthroughs described in the previous pages came easily. Marie Curie ran hundreds of experiments to isolate radium. Katherine Johnson recalculated trajectories until they were exactly right. Tu Youyou tried extraction after extraction before finding the one that worked.

In between the successes, there were setbacks. There were failed experiments, rejected applications, misread data, and — perhaps most painfully — moments when the people around them doubted whether they belonged at all.

Persistence is not the same as stubbornness. Persistence means staying committed to your question even when you need to change your approach. It means learning from what went wrong and trying again, with new information. It means choosing to see an obstacle as a problem to solve rather than a reason to stop.

This book will give you fifty opportunities to practice persistence. Some experiments will work beautifully on the first try. Others might take three or four attempts. When something doesn't work, don't close the book. Ask yourself: **What did I observe? What might have gone differently? What would I change?** Then try again.

3. Courage — Moving Forward Anyway

It takes courage to walk into a space where you're not sure you're welcome. It takes courage to ask a question when you're afraid people will think it's obvious. It takes courage to defend a finding when others tell you you're wrong. It takes courage to apply for a job, a program, or a competition when you look around and don't see many people who look like you.

Every woman in the previous pages demonstrated this kind of courage. Katherine Johnson sat down at tables where Black women were not expected to sit. Rosalind Franklin worked in a field dominated by men who often underestimated her. Mae Jemison applied to astronaut training when she was told, implicitly and sometimes explicitly, that her kind of person wasn't what NASA was looking for.

Courage, in STEM, doesn't always look dramatic. Sometimes it is as simple as raising your hand in class when you have a question. Or asking your teacher to explain something again. Or writing down a hypothesis you're not sure is right and testing it anyway.

You don't have to feel brave to act bravely. You just have to act.

4. Creativity — Seeing What Others Don't See

Ada Lovelace saw a calculating machine and imagined music. Tu Youyou read a medieval medical text and saw a malaria cure. Katherine Johnson saw numbers and saw a path to the moon.

STEM is one of the most creative disciplines there is — a fact that is not always communicated well in school. But the heart of science is imagination: the ability to look at what exists and ask, *What if it worked differently? What if I combined these two things? What if this over here is connected to that over there?*

Creativity in STEM is not about being "arty." It is about being willing to think outside the expected solution, to combine ideas from different fields, to look at an old problem from a new angle. The best scientific breakthroughs almost always involve someone seeing a connection that no one else had noticed.

As you work through the experiments in this book, practice asking creative questions.

What would happen if I changed the temperature?

What would happen if I used more?

What would happen if I did it backwards?

Let your imagination into the laboratory. It belongs there.

5. Identity — Belonging Without Changing

Perhaps the most important lesson of all: none of the women in the previous pages became scientists by becoming someone else. Marie Curie brought her Polish identity into her work, naming a new element after her home country. Mae Jemison moved between medicine, engineering, and space — refusing to be just one thing. Tu Youyou bridged two worlds of knowledge that others saw as separate.

Your identity — your culture, your background, your way of seeing the world, the specific questions your life has made you ask — is not an obstacle to your STEM career. It is an asset. Diverse perspectives produce better science. The questions you

ask, because of who you are and where you come from, may be exactly the questions the field needs.

You do not have to look like everyone else to belong in STEM.

You do not have to sound like everyone else, think like everyone else, or come from the same places everyone else comes from. The door is open. It has always been open, even when people tried to close it. Walk through it.

Now — let's do some science.

PART 1 - SCIENCE AT HOME

Science doesn't begin in a laboratory. It begins the moment you look at something familiar and ask a question you've never asked before.

Your kitchen, your garden, your bathroom shelf, your window sill — all of these are laboratories waiting to happen. The baking soda in your cupboard, the sunlight coming through your window, the lemon sitting in your fruit bowl: these are not just household objects. They are invitations.

The ten experiments in Part 1 are designed to help you discover science in the places you already know. They use materials you likely have at home or can find easily at a market. They don't require expensive equipment. They do require your curiosity, your willingness to try, and your attention.

Science at home is not "baby science." Some of the most important principles in physics, chemistry, and biology can be demonstrated with a glass of water, a handful of salt, or a jar and a candle. When you understand the science behind everyday phenomena, the world opens up in a completely new way. You start to see the physics in a soap bubble, the chemistry in a cup of tea, the biology in the bread you had for breakfast.

Pay attention. Ask questions. Write things down.

The laboratory is already yours.

IDEA 1 — THE BAKING SODA VOLCANO

Where: Kitchen table or outdoors (easier cleanup!)

Time: 20–30 minutes

Mess level: Medium — put newspaper down

What You Need:

- A plastic bottle or a cup made of clay or playdough (if you want to sculpt a volcano shape around it)
- 3–4 tablespoons of baking soda (sodium bicarbonate)
- ½ cup of white vinegar
- A few drops of red or orange food coloring
- A squirt of dish soap (optional but recommended — it makes better "lava")
- A tray to catch overflow

What To Do:

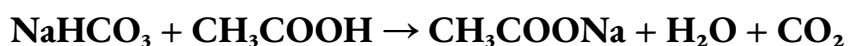
1. Build your volcano. You can use playdough or modeling clay to sculpt mountain-shaped sides around your bottle, leaving the opening at the top exposed. Or, simply place your cup on a tray without any decoration — the chemistry is the same!
2. Place the baking soda into the bottle.
3. Add the food coloring to the vinegar in a separate cup and stir.
4. If using dish soap, add a small squirt directly into the baking soda in the bottle.
5. When you're ready for the eruption, pour the colored vinegar into the bottle.
6. Watch your volcano erupt!

What's Happening?

The foamy eruption is caused by a chemical reaction between baking soda (a base) and vinegar (an acid). When they mix, they undergo an acid-base reaction that produces carbon dioxide gas (CO_2) very rapidly. The gas bubbles up through the liquid, creating the frothy, foamy "lava."

The dish soap, if you used it, traps the CO₂ bubbles inside soap film, making the foam last longer and look more dramatic.

The chemical equation looks like this:



(Sodium bicarbonate + Acetic acid → Sodium acetate + Water + Carbon dioxide)

Go Further:

- Try varying the amount of baking soda or vinegar. What changes?
- What happens if you add warm vinegar versus cold vinegar?
- Can you slow the reaction down? Can you make it faster?
- What other household acids (lemon juice, orange juice) produce a reaction with baking soda?

What You Learned:

The concepts of **acid**, **base**, **chemical reaction**, and **gas production**. You also saw that chemistry produces energy — in this case, in the form of rapidly expanding gas.

IDEA 2 — THE MINI LAVA LAMP

Where: Kitchen or desk

Time: 15–20 minutes

Mess level: Low — oil can be slippery, so work over a tray

What You Need:

- A clear glass or plastic bottle/jar (the taller, the more dramatic)
- Water
- Vegetable oil (or baby oil — clear baby oil makes the effect especially beautiful)
- Alka-Seltzer tablets OR baking soda + vinegar (one tablet per show)
- Food coloring (any color — blue or green looks especially stunning)
- A flashlight (optional, for extra effect)

What To Do:

1. Fill the container about one-quarter full with water.
2. Add several drops of food coloring to the water and stir.
3. Slowly pour vegetable oil into the container until it is about three-quarters full. Pour slowly! Watch what happens as the oil meets the water.
4. Wait until the oil and water have fully separated into distinct layers.
5. Break an Alka-Seltzer tablet into a few small pieces. Drop one piece into the container.
6. Watch what happens! The colored water will form bubbles that rise through the oil, then sink back down, again and again — like a real lava lamp.
7. Hold a flashlight under the container or shine it from the side for a beautiful glowing effect in a dim room.
8. When the bubbling stops, add another piece of tablet to start again.

What's Happening?

This experiment demonstrates three important concepts:

- **Density and Immiscibility:** Oil is less dense than water, so it floats on top. More importantly, oil and water are **immiscible** — they don't mix, because of

the way their molecules are structured. Water molecules are polar (they have a slight positive and negative charge); oil molecules are non-polar. Polar molecules are attracted to other polar molecules, and non-polar to non-polar — so the two liquids stay separate. This is why soap (which has both polar and non-polar parts) is needed to wash oily things with water.

- **The Reaction:** The Alka-Seltzer tablet contains citric acid and baking soda in dry form. When it touches water, they react to produce CO_2 gas bubbles. These bubbles attach to droplets of the colored water and lift them up through the oil. When the bubbles reach the surface, they pop, releasing the gas — and the water droplet, now heavy again, sinks back down.
- **Density in motion:** The water droplets move up when they have buoyant gas attached, and sink when the gas escapes. You're watching a continuous dance of density.

Go Further:

- What happens if you use salt water instead of plain water?

Hint: salt water is denser — how might this change things?

- Try different colors layered — can you create a three-layer lamp?
- What happens if you tilt the jar? Does the oil/water boundary stay flat?

What You Learned: Density, immiscibility, polarity, acid-base reactions, buoyancy.

IDEA 3 — THE RAINBOW IN A GLASS

Where: Near a window with direct sunlight, or any bright room

Time: 10–15 minutes

Mess level: Very low

What You Need:

Option A — Rainbow with water and sunlight

- A clear glass or bowl of water
- A small mirror
- A sheet of white paper
- Direct sunlight (morning or afternoon works best)

Option B — Rainbow with a CD or DVD

- Any old CD or DVD
- A flashlight or bright lamp
- A white wall or piece of white paper

Option C — Rainbow with a garden hose

- A garden hose with a spray nozzle
- Direct sunlight
- Your back to the sun

What To Do (Option A):

1. Fill a clear glass or bowl with water and place it near a sunlit window. The sunlight should be coming in at a low angle.
2. Hold the small mirror in the water, tilted at an angle toward the sunlight.
3. Move the mirror slowly until a rainbow appears on the wall or ceiling.
4. Hold the white paper in the path of the reflected rainbow to see it more clearly.

What To Do (Option C, outdoors):

Stand with the sun behind you. Spray water in a fine mist from your hose, aiming slightly upward and away from you. You should see a rainbow appear in the mist.

What's Happening?

Light appears to be white or colorless, but it is actually made up of all the colors of the spectrum — red, orange, yellow, green, blue, indigo, and violet (the famous ROY G BIV).

When light passes from one medium into another (like from air into water, or from water back into air), it bends. This is called **refraction**. The important thing is that different wavelengths (colors) of light bend by slightly different amounts — violet bends the most, red bends the least. This spreading out of white light into its component colors is called **dispersion**.

In your water glass experiment, the mirror reflects light through the water, which refracts it and separates the colors. In a natural rainbow, raindrops act as millions of tiny prisms, refracting and reflecting sunlight inside each drop and separating the colors before sending them back to your eyes.

The next time it rains and then the sun comes out, you will understand exactly what you're looking at.

Go Further:

- Try the glass/mirror experiment at different times of day. Does the angle of sunlight change where the rainbow appears?
- Can you use a triangular glass prism (or a clear triangular object) to separate light?
- Look up: what causes a double rainbow? How is the second rainbow different from the first?

What You Learned: Refraction, dispersion, the visible spectrum of light, wavelength.

IDEA 4 — THE FLOATING EGG

Where: Kitchen

Time: 15 minutes

Mess level: Very low

What You Need:

- 2 tall glasses or jars of similar size
- 2 raw eggs (uncracked)
- Water
- Salt (a lot — you'll need several large tablespoons)
- A spoon for stirring

What To Do:

1. Fill both glasses with water.
2. In the first glass, stir in 6–8 tablespoons of salt, adding it gradually and stirring until it fully dissolves. This takes a minute — keep going until no more salt will dissolve (the water is saturated).
3. Leave the second glass with plain fresh water.
4. Gently lower one egg into the fresh water glass. Observe what happens.
5. Gently lower the other egg into the salty water glass. Observe what happens.
6. What do you notice?

A Special Challenge — The Layered Egg: Can you make the egg float at the exact middle of the glass — not at the bottom, not at the top?

Try this: carefully fill the glass halfway with salt water (colored blue with food dye). Then very slowly pour fresh water (colored red with food dye) on top using a spoon as a guide to slow the flow. If you do it gently enough, the two layers will stay separate. Now add the egg. Where does it settle?

What's Happening?

An object floats in a liquid when its density is less than the density of the liquid. It sinks when its density is greater.

A raw egg is denser than fresh water — so it sinks. But salt water is denser than fresh water, because the dissolved salt adds mass without changing the volume very much. When enough salt is dissolved, the salt water becomes denser than the egg — so the egg floats.

This is exactly the same principle that makes the Dead Sea (the saltiest large body of water on Earth, with a salinity nearly ten times that of the ocean) famous for how easily humans float in it. Swimming in the Dead Sea, it's nearly impossible to sink. The layered challenge adds another dimension: if you layer fresh water on top of salt water, you create two layers of different density. An egg placed in this system will sink through the fresh water layer (denser than fresh water) but float on top of the salt water layer (less dense than salt water). It hangs, suspended, in the middle.

Go Further:

- What other objects float in salt water but sink in fresh water?
- What is the minimum amount of salt you need to make the egg float?
- Compare the Dead Sea to your salt solution — look up the salinity of the Dead Sea. How much salt does it contain per liter?

What You Learned: Density, buoyancy, salinity, Archimedes' principle.

IDEA 5 — CLOUD IN A JAR

Where: Kitchen (adult supervision recommended for the hot water step)

Time: 15–20 minutes

Mess level: Low

What You Need:

- A clean glass jar (a large mason jar or pickle jar works well)
- Very hot water (just boiled, but not boiling — have an adult help)
- Ice cubes
- A plate or shallow dish (to hold ice on top of the jar)
- Hairspray (optional but helps!)

What To Do:

1. Pour about 2–3 cm of very hot water into the bottom of the jar.
2. Swirl it around to warm the glass.
3. If using hairspray, give a very brief spray into the jar (just a second or two).
4. Quickly place the plate of ice cubes on top of the jar opening.
5. Wait and watch. After about 30 seconds to a minute, you should see a cloud forming inside the jar.
6. Remove the plate of ice. The cloud will drift upward and out.

What's Happening?

Real clouds form by essentially the same process you just created in miniature.

- **Evaporation:** The hot water at the bottom of the jar evaporates, sending water vapor up into the air inside the jar.
- **Cooling:** The ice on the plate cools the top of the jar. The warm, moist air rises and meets the cold air near the ice.
- **Condensation and cloud formation:** When warm, moist air cools rapidly, the water vapor condenses — turns back into tiny liquid droplets. But this condensation doesn't happen in empty air. Water molecules need tiny particles to cling to in order to form droplets. These are called *condensation nuclei*.

In nature, condensation nuclei are tiny particles of dust, pollen, or sea salt. In your experiment, the hairspray particles play this role — which is why the cloud forms better with it.

Millions of tiny water droplets suspended in the air: that's a cloud.

This same process creates the clouds in the sky. Warm, moist air rises (in weather systems, over mountains, or simply because the sun heats the ground). As it rises, it cools. When it cools enough, water vapor condenses around particles into droplets, and a cloud is born.

Go Further:

- What happens if you use very cold water instead of hot? Why doesn't a cloud form?
- Try the experiment without hairspray. What changes?
- Look up different types of clouds (cumulus, stratus, cirrus, nimbus). Can you draw them and label what weather each type brings?

What You Learned: Evaporation, condensation, cloud formation, water cycle, the role of condensation nuclei.

IDEA 6 — CAPILLARY ACTION WITH CELERY AND FLOWERS

Where: Kitchen, over 1–2 days

Time: Setup: 10 minutes. Results: 1–2 days

Mess level: Very low

What You Need:

- Several stalks of celery (with leaves still attached, if possible)
- White flowers (white carnations work best; you can also use white daisies)
- Several glasses of water
- Food coloring in different colors (red, blue, yellow)
- A knife and cutting board (ask an adult to help)

What To Do:

1. Fill three or four glasses with water. Add different food coloring to each — red to one, blue to another, etc.
2. Ask an adult to cut the bottom of each celery stalk at a fresh angle (diagonal cuts work better — more surface area).
3. Place one celery stalk in each colored water glass.
4. Do the same with the white flowers — cut the stems at an angle and place them in the colored water.
5. Leave them in a bright, warm spot and check every few hours.
6. Within a few hours, you should start to see color changes in the leaves of the celery. Within 1–2 days, the flower petals should be noticeably colored.
7. **Bonus:** If you split the stem of one flower down the middle and place each half in a different color of water, you can create a two-toned flower!

What's Happening?

This experiment demonstrates **capillary action** — one of the most important processes in biology and one of the key ways that plants transport water from their roots to their leaves.

Inside a plant stem, there are tiny tubes called **xylem vessels**. Water molecules have a property called **cohesion** — they are attracted to each other (this is why water forms droplets). They also have a property called **adhesion** — they are attracted to the walls of the xylem vessels.

These two forces, combined with the evaporation of water through the tiny pores in leaves (called **transpiration**), create a continuous upward pull. Water is literally drawn up from the roots, through the stem, all the way to the tips of the leaves — sometimes several meters in a tall tree — without any pump.

The colored water shows you the path this water takes. You can see it in the colored veins of the celery and in the petals of the flowers.

Note: Plants drink up water along with any dissolved minerals (and in your case, food dye). This is the same mechanism by which nutrients in the soil enter a plant — dissolved in water that is drawn upward through the roots.

Go Further:

- Cut across a celery stalk that has been in colored water for 24 hours. Do you see the xylem vessels? They should appear as colored rings or dots in the cross-section.
- What happens if you put celery in plain water versus sparkling (carbonated) water? Do the bubbles affect water uptake?
- Research *transpiration*: how does water leave a plant through its leaves, and how does this help pull water up from the roots?

What You Learned: Capillary action, cohesion, adhesion, transpiration, xylem vessels, plant biology.

IDEA 7 — THE LEMON BATTERY

Where: Kitchen or desk

Time: 30–45 minutes

Mess level: Low

What You Need:

- 4–6 fresh lemons (or limes — they work too)
- Copper coins or short lengths of thick copper wire
- Zinc nails or galvanized (zinc-coated) nails (find at a hardware store)
- Small LED light (from a hobby/electronics shop, or salvaged from a broken toy)
- Alligator clip wires (available in electronics or hobby stores, or you can twist wire around the nails and coins) OR regular wire with the insulation stripped off the ends
- Optional: a low-voltage voltmeter to measure the electricity produced

What To Do:

1. Roll each lemon firmly on the table to break some of the internal cells (this releases more juice inside, making the lemon more conductive).
2. Into each lemon, push one copper coin/wire and one zinc nail — about 2 cm apart. Make sure they don't touch each other inside the lemon.
3. Connect the lemons in a series (a chain): connect the zinc nail of the first lemon to the copper of the second lemon, the zinc of the second to the copper of the third, and so on.
4. Connect the zinc nail at one end of the chain and the copper piece at the other end to the two legs of your LED.
5. The LED should light up!

Troubleshooting:

- If the LED doesn't light: check that all connections are secure, make sure the zinc and copper don't touch inside the lemon, try rolling the lemons more firmly, add more lemons to the chain.
- A single lemon produces about 0.7–0.9 volts. An LED typically needs about 1.5–2 volts — so you need at least 2–3 lemons connected in series.

What's Happening?

A battery works by using chemical energy to move electrons from one place to another. The movement of electrons is what we call **electric current**.

In your lemon battery:

- The **lemon juice** (citric acid) is the **electrolyte** — the conducting liquid that allows ions to move.
- The **zinc nail** is the **negative electrode (anode)**. Zinc is more reactive than copper, so zinc atoms lose electrons readily, releasing zinc ions into the juice. The electrons left behind travel through your wire.
- The **copper coin** is the **positive electrode (cathode)**. Hydrogen ions from the lemon juice accept electrons at the copper surface.

The electrons flow from zinc → through the wire → to copper. This flow is electric current. By connecting multiple lemons in series, you add their voltages together until there's enough to power the LED.

This is essentially the same principle Alessandro Volta used in 1800 when he invented the first true battery — using discs of zinc and copper separated by cloth soaked in brine (salt water). You just made a 224-year-old invention out of a fruit bowl.

Go Further:

- Try different fruits (oranges, grapefruits, potatoes, apples). Which produces the most electricity? Why might some fruits produce more than others?
- Use a voltmeter to measure the voltage of each fruit. Compare.
- Could you power a small digital clock with enough fruit batteries?

What You Learned: Electrochemistry, electrolytes, oxidation and reduction, electric current, voltage, how batteries work.

IDEA 8 — CHROMATOGRAPHY: THE HIDDEN COLORS IN INK

Where: Kitchen or desk

Time: 20–40 minutes

Mess level: Very low

What You Need:

- Coffee filters, or white paper towels cut into strips about 2 cm wide and 15 cm long
- Water-soluble (washable) felt-tip markers in various colors — especially black, brown, and purple (avoid permanent markers)
- Pencils
- A glass or cup of water
- Tape or a pencil to hold the strips across the glass

What To Do:

1. Cut your paper into strips.
2. About 2 cm from the bottom of each strip, draw a thick dot or line with one marker color. Use a different color on each strip.
3. Tape the strip to a pencil so it hangs down. The pencil rests across the top of the glass, and the strip hangs into the water — but the ink mark should be just **above** the water line, not touching it.
4. Wait and watch. The water will slowly travel up the paper (capillary action again!), and as it passes the ink mark, it will carry the colors with it.
5. After 20–30 minutes, or when the water has almost reached the top of the strip, remove it and let it dry.
6. Look at what separated out.

What's Happening?

The colors in a marker are not always single pigments. They are mixtures of several different dye molecules — which is why this experiment is so surprising. That black

marker? It might separate into blue, red, yellow, and brown. That purple? It's probably red and blue combined.

Chromatography is a separation technique that works because different molecules travel at different speeds through the paper. The water (called the **mobile phase**) carries the dye molecules up the paper. The paper (the **stationary phase**) slows different molecules down by different amounts, depending on:

1. How strongly each dye molecule is attracted to the paper fibers
2. How easily each dye molecule dissolves in water

Molecules that cling strongly to the paper travel slowly and end up low on the strip. Molecules that dissolve easily in water travel fast and end up higher. This difference in speed separates the mixture into its component colors.

This same technique — in more sophisticated forms — is used in real laboratories for everything from analyzing food coloring to identifying drugs in a blood sample.

Note: This is also how Rosalind Franklin worked! Though her technique (X-ray crystallography) is different, chromatography is a foundational analytical chemistry tool used throughout research.

Go Further:

- Try permanent markers (you'll need rubbing alcohol as your solvent instead of water — this works for non-water-soluble inks). Compare the results.
- Try different solvents: salt water, milk, vinegar. How do the results change?
- Can you identify what colors were mixed to make a certain ink? Try to recreate the original color by mixing markers.

What You Learned: Chromatography, mixture separation, polarity, solubility, capillary action — and the fact that color is almost never what it seems.

IDEA 9 — THE SIMPLE SUNDIAL

Where: Outdoors, on a sunny day

Time: Building: 30 minutes. Using: all day!

Mess level: None

What You Need:

- A large sheet of cardboard or thick poster board
- A pencil or a straight stick (the *gnomon* — the part that casts the shadow)
- A compass (to find north)
- A clock or phone (to mark the hours)
- A ruler and permanent marker
- A sunny outdoor spot

What To Do:

1. Push the pencil or stick straight up through the center of the cardboard at a perpendicular angle (for a basic horizontal sundial). Make sure it's perfectly vertical.
2. Bring your sundial outside on a clear, sunny day and use the compass to orient it so that the cardboard faces true north (remember: a compass points to magnetic north, which is close but not exactly true north — for this experiment, it's close enough).
3. Place it in a spot that will get full sunlight throughout the day.
4. At each hour of the day (9am, 10am, 11am... and so on), go outside and use the permanent marker to trace the shadow line on the cardboard. Label it with the time.
5. At the end of the day, you have a working sundial calibrated to your location!
6. The next day, you can read the time from the shadow without a clock.

What's Happening?

A sundial works because the Earth rotates on its axis, making the sun appear to move across the sky from east to west. As the sun moves, the shadow cast by the gnomon (the stick) sweeps across the dial, pointing to different time markings.

The Earth rotates 360° in 24 hours, which means it rotates 15° every hour. This is why the shadow moves consistently and predictably.

The reason a sundial needs to be oriented correctly is that it uses the position of the sun relative to your location on Earth. Your latitude and longitude affect exactly where the shadow falls at each time of day.

Sundials are among the oldest timekeeping devices in human history. Ancient Egyptians, Greeks, and Chinese civilizations all used them. The sundial was the primary means of measuring time for thousands of years before mechanical clocks were invented in the 14th century.

What you've built is a direct connection to one of the oldest and most fundamental observations humans ever made: that time and light are connected.

Go Further:

- How accurate is your sundial? Compare it to a clock over several days.
- How does the shadow change with the seasons? If you mark the shadow length at noon each week, you can track how the sun's position in the sky changes through the year.
- Look up the **equation of time** — why does a sundial sometimes run slightly ahead or behind a mechanical clock?

What You Learned: Earth's rotation, shadows and angles, the concept of time zones, history of measurement.

IDEA 10 — MAKING SLIME: THE CHEMISTRY OF POLYMERS

Where: Kitchen or desk (protect surfaces — slime can stain)

Time: 20–30 minutes

Mess level: Medium

What You Need:

Recipe 1 (PVA Glue Slime):

- ½ cup white school glue (PVA glue)
- ½ cup liquid laundry starch (look for sodium tetraborate-based products)
OR: ½ teaspoon borax powder dissolved in 1 cup water (as the activator)
- Food coloring (optional)

Recipe 2 (Cornstarch Slime / Oobleck — no chemicals):

- 2 cups cornstarch (cornflour)
- 1 cup water
- Food coloring (optional)

What To Do (Recipe 1 — Borax Slime):

1. Pour the glue into a bowl. Add food coloring if desired.
2. If using liquid starch: gradually add the starch, stirring constantly. The mixture will begin to pull together and become less sticky. Knead with your hands for a few minutes.
3. If using borax solution: add it slowly, a tablespoon at a time, stirring well after each addition. Stop when the slime reaches your desired consistency.
4. Store in a sealed bag or container.

What To Do (Recipe 2 — Oobleck):

1. Pour cornstarch into a bowl.
2. Gradually add water, mixing with your hands. Add color if desired.

3. When mixed, it will look and act very strangely! Punch it — it feels solid. Rest your hand on it — it sinks in slowly, like liquid.

What's Happening?

PVA Slime:

PVA glue contains **polyvinyl acetate** — long chain-like molecules called **polymers**.

These chains normally slide past each other easily, which is why glue is a liquid.

When you add the borax activator, the boron compounds form **cross-links** between the polymer chains, connecting them like the rungs of a ladder. This network of cross-linked polymers creates a material that is neither fully solid nor fully liquid — it is a **non-Newtonian fluid**, specifically a **gel**.

The resulting slime stretches (the polymer chains can shift and slide when pulled slowly) but breaks if pulled quickly (because the cross-links don't have time to rearrange). It holds its shape but flows slowly under gravity.

Oobleck:

Oobleck is a **suspension** of cornstarch particles in water. Normally, the starch particles float separately. When you apply force (like punching), the particles are pushed together and briefly lock, making the material behave like a solid. When force is removed, they spread apart again and it flows like liquid.

This behavior is called **shear thickening**, and it is a property of many materials engineers study — including materials that might one day be used in flexible body armor.

The word "Oobleck" comes from a Dr. Seuss book about a strange substance that fell from the sky. Scientists kept the name.

Go Further:

- Try changing the ratio of borax to glue in Recipe 1. How does it change the texture?
- What happens when you put Oobleck in the freezer?
- Try walking on Oobleck in a container (you'll need a very large bin and a lot of cornstarch — this is a messy but extraordinary demonstration). Running on it works. Standing still on it doesn't.

- Research: what other non-Newtonian fluids exist? (Ketchup, toothpaste, quicksand, blood are all examples!)

What You Learned: Polymers, cross-linking, non-Newtonian fluids, states of matter, chemistry of everyday materials.

End of Part 1 — Science at Home

You've just explored chemistry, physics, biology, and earth science — all without leaving the house. That's what science at home looks like: not small, not simple, but accessible. Every one of these experiments connects to real scientific principles that researchers study every day in universities and laboratories around the world.

Take a moment to write in your science journal: Which experiment surprised you most? What question do you still have?

PART 2 - GIRLS IN THE SCHOOL LAB

Let's talk about something important before we begin.

For a lot of girls, the school laboratory can feel intimidating. The equipment looks complicated. The instructions sound technical. There's a pressure to get everything right on the first try, and sometimes — if you look around — you might feel like you're being watched more closely than your classmates, as if people are waiting to see whether you'll manage.

Here's what we want you to know: that feeling is common, it makes complete sense given the world we live in, and it is not an accurate reflection of whether you belong there.

Laboratory science is not a performance. It is an investigation. A good scientist is not someone who never makes mistakes — it is someone who notices mistakes carefully, understands what caused them, and uses that information. The lab is not a place where you prove you're smart. It is a place where you find things out.

The ten ideas in this part are designed to help you feel at home in the school lab — or to recreate school-lab experiences at home with simple equipment. They're organized around real scientific techniques: testing solutions with indicators, looking at cells under a microscope, separating mixtures, and studying living systems.

Some of these experiments you may have done in class already. If so, try them again with a scientist's eye — not just following the procedure, but asking: *Why this step? What would happen if I changed this? What does this result mean?*

The lab belongs to you.

IDEA 11 — TESTING pH: THE ACID-BASE SCALE IN EVERYDAY LIFE

Where: Kitchen or school lab

Time: 40–60 minutes

Mess level: Medium — use old clothes

Background: What Is pH?

pH is a scale that measures how acidic or basic (alkaline) a solution is. It runs from 0 to 14:

- **0–6:** Acidic (the closer to 0, the more acidic)
- **7:** Neutral (pure water)
- **8–14:** Basic/Alkaline (the closer to 14, the more basic)

The "p" stands for "potenz" (German for "power") and the "H" stands for hydrogen. The pH scale is a measure of the concentration of hydrogen ions (H^+) in a solution. pH matters everywhere: in the ocean, in your blood, in soil, in food preservation, in medicine, and in industrial chemistry.

What You Need:

For the natural indicator (red cabbage indicator):

- Half a head of red/purple cabbage
- Water
- A pot, a stove or kettle (ask an adult)
- A strainer

For testing:

- Small clear cups or test tubes
- Liquids to test: lemon juice, vinegar, baking soda solution, milk, tap water, orange juice, sparkling water, soap solution, tea

Making the Indicator:

1. Chop the red cabbage into small pieces.

2. Place in a pot with enough water to cover.
3. Bring to a boil, then simmer for 10 minutes. The water will turn deep purple.
4. Strain out the cabbage pieces. Let the purple liquid cool.
5. This is your indicator! Red cabbage contains a pigment called *anthocyanin* that changes color in response to pH.

What To Do:

1. Pour a small amount of the purple cabbage indicator into each cup.
2. Add a small amount of each test liquid to a different cup.
3. Stir and observe the color change.

Color Guide:

- **Red / Pink:** Very acidic (pH 1–3)
- **Red-purple:** Acidic (pH 4–5)
- **Purple (unchanged):** Neutral (pH 7)
- **Blue-green:** Mildly basic (pH 8–10)
- **Very basic:** (pH 11–14)

Record Your Results:

Make a table:

Liquid	Color change	Estimated pH	Acid or Basic?
Lemon juice			
Vinegar			
Baking soda water			
Milk			
Tap water			

What's Happening?

Anthocyanin (the pigment in red cabbage) is a **pH indicator** — a molecule that changes structure slightly depending on the concentration of hydrogen ions in its

surroundings. This structural change affects how the molecule absorbs light, which changes the color we see.

Commercial pH strips work on the same principle — they're impregnated with synthetic indicators that change color predictably at different pH levels.

Your body is extremely careful about pH. Blood must be maintained between pH 7.35 and 7.45 — a range so narrow that a deviation of even 0.1 can cause serious health problems. Your body uses a system called **buffering** to maintain this range.

Go Further:

- Can you test pH of rainwater? Is it neutral, or slightly acidic? (Hint: CO₂ in the atmosphere dissolves in rain to form carbonic acid, giving clean rainwater a pH of about 5.6. Acid rain can have pH values as low as 4 or 3.)
- What is the pH of ocean water? How is **ocean acidification** (a consequence of climate change) affecting marine life?
- Make pH strips by dipping strips of coffee filter paper in your cabbage indicator solution and letting them dry. Use them to test substances later.

What You Learned: pH, acids, bases, indicators, anthocyanin, chemistry of everyday substances.

IDEA 12 — OBSERVING ONION CELLS UNDER A MICROSCOPE

Where: School lab (microscope required; some toy microscopes work for this)

Time: 30–45 minutes

Mess level: Low

What You Need:

- A compound microscope (minimum 40x magnification)
- Microscope slides and coverslips
- A fresh onion
- Tweezers or a toothpick
- A dropper or pipette
- Water
- Iodine solution (optional, for staining — available at pharmacies)
- Scissors

What To Do:

1. Peel a dry outer layer from the onion and discard it. Break one of the inner, slightly moist layers and look for the thin, transparent membrane on the inner surface — this is the **epidermis**.
2. Carefully peel off a very small piece of this transparent membrane (1 cm × 1 cm or smaller). Use your tweezers.
3. Place the membrane flat onto the center of a clean microscope slide.
4. Add 1–2 drops of water (or iodine solution for better contrast). Iodine stains starch orange-brown and makes cell nuclei more visible.
5. Gently lower a coverslip over the specimen at an angle to avoid air bubbles.
6. Place the slide on the microscope stage and start with the lowest magnification.
7. Focus carefully, then switch to higher magnification.

What to Look For:

At 40x magnification or higher, you should be able to see:

- The rectangular outline of individual cells (onion cells have rigid cell walls that give them their geometric shape)
- The **cell wall** — the outermost boundary
- The **cell membrane** — just inside the cell wall
- The **nucleus** — a dark oval dot inside each cell (easier to see with iodine staining)
- The **vacuole** — a large central space inside each cell (takes up most of the cell's volume)

What's Happening?

Onion cells are plant cells, which means they have several features that animal cells (including human cells) don't have: a rigid **cell wall** made of cellulose (which is why they look like bricks), large central **vacuoles** that store water and waste, and **chloroplasts** — though not in this outer skin layer, which is why the cells are colorless.

The nucleus contains the cell's DNA — the complete genetic instructions for the entire onion plant. Every cell has a complete copy.

When Robert Hooke first looked at cork under a microscope in 1665, he described the tiny compartments he saw as "cells" — because they looked like the small rooms (cells) where monks lived. He had no idea he was looking at the fundamental unit of all life. You're now seeing the same thing he saw, with the same sense of wonder.

Go Further:

- Compare onion cells to other plant cells: try looking at the thin outer skin of a tomato, a leaf, or the cells of a piece of fruit.
- If you can get a prepared slide of human cheek cells, compare the animal cell to the plant cell. What's the same? What's different?
- Try making a wet mount of pond water. What living things can you find?

What You Learned: Cell structure, plant cells, cell wall, nucleus, vacuole, microscopy technique, the history of cell biology.

IDEA 13 — SEPARATING MIXTURES: FILTRATION, SEDIMENTATION, AND EVAPORATION

Where: Kitchen or school lab

Time: 45–60 minutes for all three methods

Mess level: Low to medium

Background:

A **mixture** is a combination of two or more substances that are not chemically combined — they keep their individual properties and can, in principle, be separated. Separating mixtures is one of the most important skills in chemistry. Chemists separate mixtures to get pure substances, to analyze what's in a sample, or to process materials for industry or medicine.

There are many separation methods. You're going to explore three:

Method A — Filtration (Removing Insoluble Solids)

Use when: one substance is solid and doesn't dissolve in the liquid.

Materials: sand and water mixture, a funnel, coffee filter or filter paper, a beaker or jar.

Procedures:

1. Stir sand into water to make a murky mixture.
2. Fold a piece of coffee filter paper into a cone shape and fit it into the funnel.
3. Place the funnel over the jar.
4. Slowly pour the sand-water mixture through the filter.
5. The water passes through. The sand stays on the filter.

Why it works: Filter paper has tiny pores that allow small molecules (like water) to pass through, but trap larger solid particles (like sand).

Method B — Sedimentation (Letting Gravity Work)

Use when: solid particles are heavy enough to settle over time.

Materials: muddy water or a chalk powder and water mixture, a clear jar.

Procedures:

1. Shake the muddy water mixture vigorously.
2. Place it on a shelf and leave it completely undisturbed for 30 minutes to an hour.
3. Observe how the particles settle to the bottom, leaving clearer water above.

Why it works: Gravity pulls denser particles downward. Given time, they settle. This is called sedimentation. Pouring the clear top liquid carefully off is called **decanting**.

Method C — Evaporation (Recovering Dissolved Solids)

Use when: you have a dissolved solid and want to recover it — like getting salt back from salt water.

Materials: salt water solution, a shallow dish or tray.

Procedures:

1. Pour salty water into a wide, shallow dish.
2. Leave in a warm, sunny place (or gently heat on a stove with adult supervision).
3. Wait for the water to evaporate.
4. As the water disappears, you'll see salt crystals forming.

Why it works: Water molecules evaporate (turn into gas and escape into the air) when heated. The salt molecules, too heavy to evaporate, are left behind. This is also how sea salt is harvested in coastal salt pans around the world.

Connecting to the Real World:

- **Water treatment plants** use filtration, sedimentation, and chemical treatment to clean water.
- **Pharmaceutical companies** use sophisticated versions of these techniques to isolate pure drug compounds.
- **Mining operations** separate valuable minerals from rock using density differences and chemical extraction.
- **Coffee and tea** are prepared using filtration every day — the filter keeps the grounds back while the liquid passes through.

Go Further:

- Can you design a more effective sand filter? (Try layering sand, gravel, and charcoal in a bottle with holes poked in the bottom — this mimics a basic water filter.)
- What would you do if you needed to separate salt AND sand from water? (You'd need multiple steps. Think it through.)
- Research how communities without access to clean water use available materials to filter and purify water.

What You Learned: Filtration, sedimentation, evaporation, mixtures, separation techniques, real-world applications.

IDEA 14 — ACID-BASE REACTIONS: PRODUCING GAS AND NEUTRALIZATION

Where: Kitchen or school lab

Time: 30–40 minutes

Mess level: Medium

What You Need:

- Baking soda (sodium bicarbonate) — a base
- White vinegar, lemon juice, or citric acid solution — acids
- Baking powder (a mixture of baking soda and a solid acid)
- Clear cups or test tubes
- Balloons
- Small plastic bottles
- Measuring spoons

Experiment A — Gas Trapping:

1. Add 2 tablespoons of baking soda to a small plastic bottle.
2. Stretch the neck of a balloon over a separate container.

3. Pour 3 tablespoons of vinegar into the balloon using a funnel.
4. Fit the balloon neck over the bottle opening, keeping the vinegar in the balloon.
5. When ready, tip the balloon upright so the vinegar falls into the bottle.
6. The balloon will inflate as CO_2 is produced!

This demonstrates: The acid-base reaction produces CO_2 gas, and you can collect and measure it. Baking uses this principle — baking powder in a cake produces CO_2 when it heats up, making the cake rise.

Experiment B — Neutralization:

Neutralization happens when an acid and a base react exactly to produce a neutral salt and water.

1. Make a very dilute baking soda solution ($\frac{1}{4}$ teaspoon in 1 cup of water). This is your base.
2. Make a red cabbage indicator solution (from Idea 11).
3. Add 1 tablespoon of the baking soda solution to a cup and add indicator. Note the color (it should be blue-green: basic).
4. Very slowly add drops of vinegar, stirring after each. Watch the color change.
5. At the exact neutral point, the indicator will return to its original purple.
6. If you add too much vinegar, it will turn pink/red.

The equation: NaHCO_3 (baking soda) + CH_3COOH (acetic acid/vinegar) $\rightarrow$ CH_3COONa (sodium acetate, a neutral salt) + H_2O + CO_2

You are watching the chemistry of **antacids** happen in real time. When someone has heartburn (too much stomach acid), they take an antacid — which is a base that neutralizes the excess acid in the stomach.

What You Learned: Acid-base chemistry, neutralization, gas production, real-world applications (antacids, baking, fermentation).

IDEA 15 — STARCH TESTING IN FOOD: THE IODINE TEST

Where: Kitchen or school lab

Time: 30 minutes

Mess level: Low (iodine stains — use old cloth)

Background:

Starch is a **polysaccharide** — a large carbohydrate molecule made of long chains of glucose. It's one of the most important energy storage molecules in plants. When you eat foods containing starch (bread, pasta, rice, potatoes), your digestive system breaks the starch back down into glucose, which your cells use for energy.

Iodine solution (iodine dissolved in potassium iodide) turns from amber-brown to dark blue-black in the presence of starch. This is because the iodine molecules fit inside the spiral structure of starch molecules, and this changes how they absorb light.

What You Need:

- Iodine solution (available at pharmacies — ask for iodine antiseptic)
- A dropper or pipette
- A white plate or white paper
- Foods to test: bread, cracker, potato, apple, onion, rice (cooked), pasta, cheese, egg white, sugar, flour

What To Do:

1. Place small samples of each food on your plate.
2. Add 2–3 drops of iodine solution to each sample.
3. Observe and record the color: Does it turn blue-black (starch present)? Or stay amber-brown (no starch)?

Make a Prediction First:

Before you test, predict which foods you think contain starch. Write it down. Then see how your predictions compare to the results.

This is one of the most important habits in science: **predicting before observing**. A prediction based on prior knowledge and reasoning is called a **hypothesis**. Whether it's right or wrong, you learn from testing it.

Expected Results:

- **High starch:** bread, crackers, potato, cooked pasta, rice, flour
- **Low or no starch:** apple, onion, cheese, egg white, sugar (note: sugar is a simple carbohydrate, not a polysaccharide — iodine does not react with it)

Going Deeper:

Saliva contains an enzyme called **amylase** that begins breaking down starch in your mouth. You can test this:

1. Chew a plain cracker for 2–3 minutes without swallowing.
2. Notice that it starts to taste slightly sweet — this is the starch being converted to sugar by your salivary amylase.
3. Spit a small amount onto a test surface and add iodine.
4. Compare to an un-chewed cracker. Is there less starch in the chewed sample?

What You Learned: Starch, polysaccharides, carbohydrates, iodine indicator, enzyme action, digestion, macronutrients.

IDEA 16 — INVESTIGATING HOW TEMPERATURE AFFECTS DISSOLVING SPEED

Where: Kitchen or school lab

Time: 40–50 minutes

Mess level: Low

This is a proper controlled experiment — the kind scientists run every day.

A **controlled experiment** tests one variable at a time while keeping everything else the same. This is how you can be confident that your results are caused by the thing you changed, and not by something else.

In this experiment, your variable is **temperature**. Everything else — the type of solute, the amount of solute, the size of the crystals, how you stir — should stay the same.

Your Question: Does temperature affect how quickly sugar (or salt) dissolves in water?

Your Hypothesis: Write down your prediction before you start.

Variables:

- **Independent variable** (what you change): temperature of water (cold, room temperature, hot)
- **Dependent variable** (what you measure): time taken for the sugar to dissolve
- **Controlled variables** (what stays the same): type and amount of sugar, amount of water, stirring rate, size of sugar crystals

What You Need:

- 3 identical cups or beakers
- Cold water, room temperature water, hot water (just boiled — ask an adult)

- A thermometer (recommended, to measure the exact temperature of each)
- 3 equal portions of granulated sugar (or salt): 2 teaspoons each
- A stopwatch or phone timer
- A spoon for stirring (stir the same number of times per second for each trial)

What To Do:

1. Measure and record the temperature of each water sample.
2. Pour equal amounts of each water temperature into separate cups.
3. At the same moment (or as close as possible), add one portion of sugar to each cup.
4. Stir each at the same rate. Start your timer.
5. Record how long it takes for the sugar to fully dissolve in each cup.
6. Repeat the experiment 2–3 times to make sure your results are consistent (this is called *replication*).

Record Your Data:

Trial	Cold Water Temp	Time	Room Temp	Time	Hot Temp	Time
1						
2						
3						
Average						

What's Happening?

Dissolving happens when solvent molecules (water) break apart and surround solute molecules (sugar). At higher temperatures, the water molecules move faster (they have more kinetic energy). This means:

1. They collide more frequently with sugar molecules.
2. They break apart the sugar crystal structure more quickly.
3. They surround sugar molecules more effectively.

The result: hot water dissolves sugar significantly faster than cold water.

This has practical applications everywhere: hot tea dissolves sugar quickly; cold drinks need stirring much longer; industrial chemical processes often use heat to speed up dissolving.

What You Learned: Dissolving, solubility, temperature effects, experimental design, variables, data collection, replication.

IDEA 17 — THE BALLOON LUNG MODEL

Where: Home or school lab

Time: 30–40 minutes to build; use it for ongoing demos

Mess level: Low

What You Need:

- A clear plastic bottle (1–2 liters), with the bottom cut off
- 2 small balloons (for the lungs)
- 1 large balloon (cut in half crossways — for the diaphragm)
- 2 short straws (or a Y-shaped plastic connector)
- Tape and scissors
- Plasticine or modeling clay (to seal gaps)

What To Do:

1. Cut the bottom off the plastic bottle. This is your chest cavity.
2. Connect the two small balloons to the two ends of a Y-shaped straw or two straws joined together. These represent the lungs connected to the windpipe (trachea).
3. Insert the straw end up through the bottle cap and seal around it with clay. The small balloons should hang inside the bottle.
4. Take the large balloon, cut off the neck, and stretch the remaining dome over the open bottom of the bottle. Secure with tape or a rubber band. This represents the diaphragm.
5. Try pulling down on the bottom balloon (the diaphragm) while watching the small balloons inside. They should inflate.
6. Push the bottom balloon up. The small balloons deflate.

What's Happening?

Your model demonstrates the mechanics of breathing:

When you breathe in (**inhalation**):

- Your diaphragm (a dome-shaped muscle below your lungs) contracts and flattens, pulling downward.
- This increases the volume of the chest cavity.
- Increasing volume decreases pressure (Boyle's Law: pressure and volume are inversely related at constant temperature).
- Air rushes from higher pressure outside to lower pressure inside your lungs.

When you breathe out (**exhalation**):

- Your diaphragm relaxes and moves upward (dome shape again).
- This decreases chest cavity volume, increasing pressure.
- Air moves from higher pressure inside to lower pressure outside.

The model shows this beautifully: pulling the bottom balloon down represents the diaphragm flattening, which inflates the "lungs" inside. It's the exact same principle — pressure difference driving airflow.

What You Learned: Respiratory system, diaphragm, lung mechanics, Boyle's Law, pressure and volume.

IDEA 18 — MEASURING HEART RATE BEFORE AND AFTER EXERCISE

Where: Home or school (needs a bit of space to exercise)

Time: 30–40 minutes

Mess level: None

What You Need:

A stopwatch or phone with a timer

Your fingers and a pulse point (wrist or neck)

A notebook and pen

Finding Your Pulse:

1. Place two fingers (index and middle) gently on the inside of your wrist, just below the base of your thumb. Or on the side of your neck, below the jaw. Don't press too hard — a light touch is enough.
2. Count the beats for 15 seconds. Multiply by 4. This gives you your heart rate in beats per minute (BPM).

The Experiment:

1. **Resting rate:** Sit quietly for 2 minutes, then measure your heart rate. This is your *resting heart rate*. Record it.
2. **After light exercise:** Do 2 minutes of slow walking in place. Measure your heart rate immediately. Record.
3. **After moderate exercise:** Do 2 minutes of brisk jogging in place or jumping jacks. Measure immediately. Record.
4. **After intense exercise:** Do 2 minutes of sprinting in place as fast as you can. Measure immediately. Record.
5. **Recovery:** Sit still and measure your heart rate every 2 minutes for 10 minutes. How long does it take to return to your resting rate?

Your Data Table:

Condition	Heart rate (BPM)
Resting	
After light exercise	
After moderate exercise	
After intense exercise	
2 min recovery	
4 min recovery	
6 min recovery	
8 min recovery	
10 min recovery	

What's Happening?

Your heart pumps oxygenated blood to your muscles and organs. During exercise, your muscles need more oxygen to produce energy — so your heart beats faster to deliver more blood.

The amount of blood your heart pumps per minute is:

$$\text{Cardiac Output} = \text{Heart Rate} \times \text{Stroke Volume}$$

(Stroke volume is the amount of blood pumped per beat — typically about 70 ml at rest.)

At rest:

$$\sim 70 \text{ BPM} \times 70 \text{ ml} = \sim 4,900 \text{ ml/min (about 5 liters)}$$

During intense exercise:

$$\text{up to } 200 \text{ BPM} \times 120 \text{ ml} = \sim 24,000 \text{ ml/min!}$$

Recovery time (how long it takes your heart rate to return to normal) is an indicator of cardiovascular fitness. Athletes generally recover faster because their hearts are stronger and more efficient.

Go Further:

- Compare resting heart rates among classmates or family members. Are there differences? What factors might influence resting heart rate?
- What is the maximum heart rate? How is it estimated? (Common formula: $220 - \text{your age}$)
- Research: why do trained athletes sometimes have resting heart rates as low as 40–50 BPM, while the average is 60–100?

What You Learned: Heart rate, cardiovascular physiology, exercise and the body, cardiac output, recovery rate, experimental data collection.

IDEA 19 — MODELING THE HUMAN CIRCULATORY SYSTEM

Where: Desk or table

Time: 45–60 minutes

Mess level: Low

What You Need:

A large sheet of poster paper or flip chart paper

Red and blue colored pens or markers

A basic diagram of the human body outline (you can trace your own or print one)

Reference: a biology textbook or trusted internet image of the circulatory system

What To Do:

1. Draw (or trace) a simple outline of the human body on your poster paper.
2. Draw the heart in the center of the chest — slightly to the left.
3. Using **red** for oxygenated blood and **blue** for deoxygenated blood, draw the major blood vessels:
 - The **aorta** (main artery leaving the heart, carrying oxygenated blood to the body)
 - The **pulmonary artery** (carrying deoxygenated blood from the heart to the lungs)
 - The **pulmonary veins** (carrying oxygenated blood from the lungs back to the heart)
 - The **vena cava** (carrying deoxygenated blood from the body back to the heart)
 - Major arteries to the head, arms, and legs
 - Major veins returning from the head, arms, and legs
4. Label all the structures.
5. Draw arrows to show the direction of blood flow.
6. In a corner of the poster, draw a simple diagram of the **four chambers** of the heart:

- Right atrium and right ventricle (receive and pump deoxygenated blood)
- Left atrium and left ventricle (receive and pump oxygenated blood)

Understanding the Two Loops: The circulatory system is actually two connected loops:

- **Pulmonary circulation:** Heart → lungs → heart
Deoxygenated blood leaves the right ventricle, travels to the lungs (where it picks up oxygen and releases CO₂), then returns to the left atrium.
- **Systemic circulation:** Heart → body → heart
Oxygenated blood leaves the left ventricle through the aorta, travels to all organs and tissues (delivering oxygen, picking up CO₂), then returns to the right atrium.

The Heart as a Pump:

The heart is a **double pump**. The right side pumps blood to the lungs; the left side pumps blood to the rest of the body. The heart has four chambers (atria on top, ventricles on the bottom) and four valves that ensure blood flows in only one direction.

In an average lifetime, the human heart beats approximately **2.5 billion times** without stopping.

What You Learned: Circulatory system, heart anatomy, arteries and veins, oxygenated/deoxygenated blood, pulmonary and systemic circulation.

IDEA 20 — INVESTIGATING HOW LIGHT AFFECTS PLANT GROWTH

Where: Home — needs a sunny and a dark/shaded spot

Time: Setup 20 minutes; observations over 2–4 weeks

Mess level: Very low

This is a long-term experiment. The setup takes minutes, but the real science happens over weeks as you observe and record changes.

What You Need:

- 6–8 identical small pots or yogurt cups with holes poked in the bottom
- Potting soil
- Seeds of the same type (beans, cress, sunflowers, or lettuce work well — fast-growing ones are best)
- Water
- Labels
- A notebook for daily observations

Setting Up:

1. Fill all pots with the same amount of soil.
2. Plant the same number of seeds (3–4 per pot) at the same depth in all pots.
3. Water all pots equally.
4. Place the pots in three different light conditions:
 - Group A (Full light): Sunny windowsill or outdoors
 - Group B (Partial light): Indoors away from the window, some indirect light
 - Group C (No light): Inside a cupboard or box — completely dark
5. Water all pots equally every 2 days. Keep all other conditions (temperature, soil, water amount) as identical as possible.

Observe and Record Daily:

For each group, record:

- Date
- Height of seedlings (in cm)
- Color of stems and leaves (green, pale yellow, white?)
- General health (strong and upright? Thin and droopy?)
- Any other observations (roots showing? Leaves curling?)

Draw simple sketches of each group's plants every few days.

What to Expect:

- **Full light plants:** Will be green, compact, sturdy, healthy.
- **Partial light plants:** Will grow somewhat normally, but may be slightly paler and taller (reaching for light).
- **No light plants:** Will sprout (they have stored energy in the seed) but will be **etiolated** — pale yellow or white, very tall and thin, with small or no leaves. They will eventually die without light.

What's Happening?

Plants make their own food through **photosynthesis** — using light energy, water, and carbon dioxide to produce glucose and oxygen. The green color of plants comes from **chlorophyll**, the pigment that absorbs light.

Without light, plants cannot photosynthesize. They use the stored energy in the seed at first (this is why dark-grown plants can still sprout), but once that's used up, there is no energy source for growth.

Etiolation (the pale, stretched growth in darkness) is actually an adaptation — the plant is literally stretching upward as fast as possible, trying to reach light. This is a programmed response to being in darkness, controlled by plant hormones.

What You Learned: Photosynthesis, chlorophyll, plant growth, light as an energy source, etiolation, controlled experiments, long-term data collection.

End of Part 2 — Girls in the School Lab

Look back at the ten experiments in Part 2. You've investigated pH, looked at living cells, separated mixtures, tested food for starch, run a controlled dissolving experiment, built a lung model, measured your own heart rate, mapped the circulatory system, and tracked plant growth over weeks.

These are real laboratory skills. The techniques you've practiced — making hypotheses, controlling variables, recording data, and drawing conclusions — are the same ones used in research laboratories around the world every day.

Take a moment to ask: **Which experiment changed the way you see your own body? Which one made you want to learn more?**

PART 3 - ENGINEERING AND DESIGN FOR EVERYDAY LIFE

Engineering is not just about machines and factories. It is the art of solving problems — using creativity, materials, and systematic thinking to build things that work.

Every object in your life is the product of engineering: the chair you sit on, the bag you carry to school, the bridge you cross on the way to the market, the water that comes from your tap. Someone looked at a problem — *"people need to cross this river," "water needs to get to people's homes," "students need something to carry their books"* — and designed a solution.

The experiments in Part 3 invite you to think like an engineer. This means thinking not just about what you're building, but why you're building it, what constraints you're working under, and how you will test whether your design works.

Engineers use something called the **Design Process**:

1. **Identify the problem.** What exactly are you trying to solve?
2. **Research.** What do you already know? What can you learn?
3. **Brainstorm solutions.** Think of as many ideas as possible without judging them.
4. **Choose a design.** Pick one idea and plan it out.
5. **Build it.** Make your design with the materials available.
6. **Test it.** Does it do what it's supposed to do?
7. **Evaluate and improve.** What worked? What didn't? How can you make it better?
8. **Share your results.** Communication is part of engineering.

These steps aren't always linear — you'll often loop back and redesign. That's not failure. That's engineering.

IDEA 21 — BUILDING A BRIDGE FROM POPSICLE STICKS

Where: Table or desk

Time: 1–2 hours

Mess level: Low (some glue)

The Challenge: Design and build a bridge using only popsicle sticks and white glue that can:

- Span a gap of at least 30 cm (place it between two books or stacks)
- Hold as much weight as possible

What You Need:

- Popsicle sticks / craft sticks (50–100)
- White glue or craft glue
- Binder clips or clothes pegs (to hold pieces while glue dries)
- 2 books or boxes to serve as bridge supports
- Coins or small weights for testing
- A ruler

Engineering Concepts to Think About:

Before you build, learn about the forces at work:

- **Compression:** Forces that push or squeeze. When you stand on a bridge, gravity pushes down — the bridge deck is under compression.
- **Tension:** Forces that pull or stretch. In a suspension bridge, the cables are under tension — they're being stretched.
- **Triangles are strong!** If you look at bridges, trusses, and towers, you'll see triangles everywhere. A triangle is the only polygon that cannot be deformed without changing the length of its sides. Squares and rectangles can be pushed into parallelograms — triangles cannot. This makes them the engineer's best friend.

Design Phase:

Sketch your bridge before building. Consider:

- How will you distribute weight across the bridge?
- Will you use a truss design? (Rows of triangles.)
- How will the bridge connect to its supports?
- How can you strengthen the roadway (flat surface)?

Look at real bridge designs online for inspiration: beam bridges, arch bridges, truss bridges, suspension bridges. Each handles forces differently.

Building Phase:

Build your bridge according to your design, letting glue dry fully at each stage (or use binder clips to hold joints while they dry — this takes patience, but it matters!).

Tip: A bridge built with triangular trusses on its sides will outperform a simple flat bridge of the same materials.

Testing Phase:

Place the bridge across the gap. Load coins or weights onto the center (the weakest point). Record how much weight it holds before it fails. Observe where it fails first — this tells you where your design was weakest.

Go Further:

- Research the Forth Bridge in Scotland or the Millau Viaduct in France — these are engineering marvels. What structural principles do they use?
- Can you redesign your bridge after seeing where it failed?

What You Learned: Structural engineering, compression and tension, triangles in structures, design process, bridge types.

IDEA 22 — DESIGNING A TOWER FROM PAPER THAT HOLDS WEIGHT

Where: Table or desk

Time: 30–45 minutes

Mess level: Low

The Challenge: Using only **10 sheets of regular A4 paper** and **30 cm of tape**, build the tallest tower you can that can support the weight of a hardcover book on top for at least 10 seconds.

Rules:

- Only 10 sheets of paper (you can fold, roll, cut, curl, or layer them)
- Maximum 30 cm of tape total
- Must be freestanding (no leaning against walls)
- Must support a book on top for 10 seconds

Think Before You Build:

Paper seems flimsy — but it can be remarkably strong when used correctly. A flat sheet collapses easily under vertical load. But roll that same sheet into a cylinder, and it can support many times its own weight. This is because the rolled shape distributes force evenly around the circumference.

Consider:

- Columns (rolled paper tubes) can bear vertical loads
- A wide base gives stability
- Cross-bracing between columns prevents toppling
- A flat paper platform on top gives the book a stable resting surface

Build, Test, Revise:

Build your design, then test it. If it falls, look at where and why. Redesign. Try again. In competitive engineering challenges, teams often build multiple versions — and the later versions are almost always better.

Real-World Connection:

Tall buildings use similar principles: a strong, wide base; columns to carry load; cross-bracing or triangulated frames to resist sideways forces (like wind). The Eiffel Tower, for example, was designed as a lattice structure specifically to resist the wind that would have toppled a solid column of the same height.

What You Learned: Structural load, compression, columns, design iteration, engineering under constraints.

IDEA 23 — THE BALLOON-POWERED CAR

Where: Smooth floor (hardwood or tile works best)

Time: 45–60 minutes to build and test

Mess level: Low

The Challenge: Build a vehicle that moves using only the power of a balloon. The car that travels the farthest wins.

What You Need:

- A round balloon (longer sausage-shaped balloons actually work better)
- A flexible straw (or two straws taped together)
- A base: a piece of cardboard, an old plastic bottle, a small wooden board
- Wheels: bottle caps, spools, slices of cardboard tubes, small plastic wheels from a toy kit
- Skewers or pencils (as axles)
- Tape, scissors, push-pins or thumbtacks
- A smooth floor for testing

The Science:

Newton's Third Law of Motion: **For every action, there is an equal and opposite reaction.**

When you blow up the balloon and release it, air rushes out of the nozzle backward. The reaction to this force propels the car forward. This is the same principle that powers jet engines and rockets.

For your car to travel far, it needs:

- **Low friction:** Wheels that spin freely; a smooth floor; minimal weight
- **Efficient thrust:** A straight nozzle so air exits in one direction; a larger balloon for more air
- **Stability:** Balanced weight distribution so the car doesn't spin

Building Tips:

1. Thread your straw through a small hole in the balloon neck so air can be blown in through the straw.
2. Attach the straw (horizontally) along the underside or top of your vehicle base, pointing toward the back.
3. Attach your axles (skewers) through the base, with wheels at each end.
4. Make sure wheels are aligned and spin freely.
5. Blow up the balloon through the straw, pinch the straw shut, set the car down, and release.

Test and Improve:

- Does the car go straight? (If not, the thrust nozzle may not be aligned)
- How far does it travel?
- What changes improve the distance?

What You Learned: Newton's Third Law, thrust and propulsion, friction, engineering design, wheel and axle systems.*

IDEA 24 — THE MINI WINDMILL

Where: Outdoors or near a fan indoors

Time: 45–60 minutes

Mess level: Low

The Challenge: Build a small windmill (wind turbine) that spins efficiently in a breeze and can lift a small object (like a paper clip on a thread) as it spins.

What You Need:

- A cork, styrofoam ball, or small ball of clay (for the hub)
- Popsicle sticks or stiff cardboard (for blades)
- A wooden skewer or pencil (as the axle)
- Thread or thin string
- A small paper clip or eraser (to lift)
- Tape and scissors
- A fan (to test indoors)

Design Considerations:

The angle of the blades matters enormously. Windmill blades are not flat — they are angled (like a propeller) so that when wind hits them, they produce rotational force. The angle of attack — the angle at which the blade meets the wind — determines efficiency.

Try different blade angles and observe which makes the windmill spin faster. Also consider the number of blades (most wind turbines have 3) and blade shape.

The Science:

Wind turbines convert **kinetic energy** (energy of moving air) into **rotational mechanical energy** (spinning motion) and ultimately into **electrical energy** when a generator is attached.

The amount of energy available in wind increases with the **cube** of the wind speed. This means doubling the wind speed gives you 8 times more energy. This is why

offshore wind farms (where winds are consistently strong) produce much more energy than inland ones.

Wind energy is one of the fastest-growing renewable energy sources in the world. In many countries, it now generates electricity at lower cost than coal or gas.

What You Learned: Kinetic energy, wind turbine principles, blade design, angle of attack, renewable energy.

IDEA 25 — THE SIMPLE WATER FILTER

Where: Outdoors or kitchen sink (messy water involved)

Time: 45–60 minutes to build; testing takes 20 more

Mess level: Medium

The Challenge: Design and build a multi-layer water filter using natural materials that can take visibly muddy water and produce clearer, cleaner water.

Important note!! Your filter will NOT produce water that is safe to drink. This is about understanding filtration principles — not about creating drinkable water, which requires additional disinfection (boiling, UV treatment, chemical treatment).

What You Need:

- A large plastic bottle, cut in half
- Gravel or small pebbles
- Coarse sand
- Fine sand
- Activated charcoal (fish tank charcoal, from a pet store)
- Cotton balls or a piece of cloth
- A second container to collect filtered water
- Muddy water (mix soil into water)

Layering Your Filter (from top to bottom in the upturned bottle half):

1. Cotton ball or cloth (first layer — catches large debris)
2. Fine sand (5 cm)
3. Coarse sand (5 cm)
4. Gravel (5 cm)
5. Activated charcoal (3–4 cm) — this removes some chemical impurities and odors

Pour muddy water in at the top and collect the filtered water in a container below.

The Science: Real water treatment uses the same principles, scaled up enormously.

1. **Screening** — removes large debris
2. **Sedimentation** — allows particles to settle
3. **Filtration** — passes water through sand and gravel
4. **Activated carbon filtration** — removes organic chemicals, chlorine, some metals
5. **Disinfection** — kills bacteria and viruses (using chlorine, UV, or ozone)

Access to clean water is one of the greatest global challenges. Over 2 billion people lack access to safe drinking water. Engineers, chemists, and public health scientists work together to solve this problem.

What You Learned: Water filtration, multi-stage treatment, activated charcoal, real-world water systems, global water challenges.

IDEA 26 — DESIGNING A HEAT-RESISTANT HOUSE FROM RECYCLED MATERIALS

Where: Home

Time: 1–2 hours to build; testing takes 30 minutes

Mess level: Low to medium

The Challenge: Using only recycled and everyday materials, build a small model house that keeps its interior as cool as possible when placed under a lamp or in direct sunlight. The house with the smallest temperature rise inside wins.

What You Need:

- Recycled cardboard boxes (various sizes)
- Newspaper, bubble wrap, aluminum foil, cotton fabric, styrofoam pieces
- Scissors, tape, glue
- 2 thermometers (or 1 that you alternate)
- A lamp or a sunny window

Design Principles to Consider:

- **Color matters:** Light colors reflect heat; dark colors absorb it. White roofs stay cooler.
- **Insulation:** Trapping air in small pockets slows heat transfer. Bubble wrap, crumpled newspaper, and styrofoam all work by this principle.
- **Reflectivity:** Aluminum foil reflects radiant heat.
- **Air circulation:** Good ventilation (small openings in the right places) allows hot air to escape and cooler air to enter.

Test Your Design:

1. Place your house model under a lamp or in the sun.
2. Record the starting temperature inside.
3. After 15 minutes, record again. How much has it risen?
4. Compare to an uninsulated cardboard box as a control.

Real-World Connection:

In many parts of the world, cooling buildings is one of the largest uses of energy — and as climate change brings higher temperatures, this is only increasing. Passive cooling (using design rather than air conditioning) is a growing field in sustainable architecture. Traditional buildings in many cultures already incorporate these principles: thick mud walls in the Middle East, elevated houses in Southeast Asia, courtyard designs that channel air — these are all forms of passive thermal design developed over centuries.

What You Learned: Thermal insulation, heat transfer (conduction, convection, radiation), passive cooling, sustainable design, materials science.

IDEA 27 — THE SUSPENSION BRIDGE MODEL

Where: Table

Time: 60–90 minutes

Mess level: Low

The Challenge: Build a model suspension bridge using string, popsicle sticks, and tape that can span 40 cm and hold weight.

Key Concept: In a suspension bridge, the roadway hangs from cables (suspended). The cables are under **tension** — they're being pulled. The towers are under **compression** — they're being squashed downward by the cable's pull. By hanging the roadway, you distribute the weight across the long cables, which allows suspension bridges to span much longer distances than beam bridges.

Famous examples: the Golden Gate Bridge (San Francisco), the Forth Road Bridge (Scotland), the Jiaxing-Shaoxing Sea Bridge (China).

Build It:

1. Build two towers from popsicle sticks (layered and glued) — approximately 15 cm tall.
2. Run two main cable strings from one tower to the other, looping over the tops and anchoring at both ends.
3. Hang vertical "suspender" strings down from the main cable at regular intervals.
4. Attach a roadway of popsicle sticks to the bottom of the suspenders.
5. Test with increasing weight.

What You Learned: Tension, compression, load distribution, suspension bridge design, long-span engineering.

IDEA 28 — THE MINI CATAPULT

Where: Outdoors or large indoor space

Time: 45–60 minutes

Mess level: Low

The Challenge: Build a working catapult from popsicle sticks, a rubber band, and a spoon that launches a marshmallow as far as possible.

What You Need:

7–10 popsicle sticks

Rubber bands

A plastic spoon

Marshmallows or small balls of paper as projectiles

Build It:

1. Stack 5 popsicle sticks and bind them tightly together with rubber bands at both ends — this is your base platform.
2. Take 2 more sticks and stand them vertically, one on each side of the platform, about one-third from one end. Bind them to the platform.
3. Lay one more stick across the two vertical ones (the launch arm pivot point) and secure loosely with a rubber band so it can rotate.
4. Attach the plastic spoon to one end of the launch arm. Connect a rubber band from the other end to the base platform (to create tension).
5. Load a marshmallow in the spoon, pull down and release.

The Science:

A catapult stores **potential energy** (in the stretched rubber band) and converts it to **kinetic energy** (projectile motion). Once launched, the marshmallow follows a curved path called a **parabola** — determined by its initial speed, launch angle, and gravity.

Projectile motion equations describe this path precisely. The optimal angle for maximum distance is 45° .

What You Learned: Potential and kinetic energy, projectile motion, parabola, simple machines, lever mechanics.

IDEA 29 — THE SIMPLE PULLEY MACHINE

Where: Home or school

Time: 30–45 minutes

Mess level: Very low

The Challenge: Build a simple pulley system and demonstrate how it reduces the effort needed to lift a load.

What You Need:

- A spool of thread or a small wheel with a groove (or use a metal binder ring)
- String or thin rope
- A hook or pencil to mount the pulley from
- Weights (books, cans, bags of rice)
- A spring scale or kitchen scale (to measure force)

Build It:

Single fixed pulley:

1. Hang your spool/ring from a hook or over a door edge.
2. Run string over the groove.
3. Attach a weight to one end.
4. Pull down on the other end to lift the weight.

With a fixed pulley, the effort equals the load — but the direction of force changes (you pull down to lift up). This is useful when pulling upward is awkward.

Movable pulley (mechanical advantage = 2):

1. Attach one end of the string to the fixed point.
2. Run string around a movable pulley attached to the weight.
3. Pull the free end of the string.

Now you only need half the force to lift the weight — but you have to pull twice the distance.

This is a **mechanical advantage** — a machine lets you trade force for distance (or distance for force).

What You Learned: Simple machines, pulley, mechanical advantage, force and distance trade-off, work.

IDEA 30 — THE EGG DROP CHALLENGE

Where: Outdoors (or indoors with a high ceiling and forgiving floors)

Time: 60–90 minutes to design and build; 10 minutes to test

Mess level: Potentially very high — have paper towels ready

Recommended: Get your friends to join you!! The more the merrier.

The Ultimate Engineering Challenge!! Design a device using only the allowed materials that will protect a raw egg from breaking when dropped from a height of at least 2 meters.

Allowed Materials (pick a budget, e.g., only 20 items total from this list):

- Straws
- Rubber bands
- Tape
- Cotton balls
- Bubble wrap
- Plastic bags
- Newspaper
- Cardboard
- Popsicle sticks
- String
- Sponge pieces
- Foam
- Small plastic bag

Rules:

- The egg must be raw
- The device must be dropped (not thrown)
- The drop height is at least 2 meters (measure carefully)
- If the egg survives, increase the height

Design Thinking:

The physics challenge: When the device hits the ground, it decelerates rapidly. The egg inside experiences force ($F = ma$ — the greater the deceleration, the greater the force). Your job is to **extend the time of deceleration** to reduce the peak force on the egg.

Strategies:

- **Crumple zone:** Soft, compressible material around the egg absorbs impact by deforming slowly.
- **Suspension:** Suspending the egg inside the device so it doesn't hit the walls directly on impact.
- **Drag:** A parachute slows descent, reducing impact speed.
- **Distribute impact:** Multiple contact points spread the force.

The best solutions often combine multiple strategies.

Real-world equivalent: car safety engineering (crumple zones, airbags, seat belts all serve similar functions as your egg protection device).

After testing, examine failures. *Where did the egg crack? What does that tell you about where your design was weak?*

What You Learned: Force, deceleration, impulse-momentum theorem, crumple zones, engineering design iteration, creative problem-solving.

End of Part 3 — Engineering and Design for Everyday Life

Look around you right now. Everything you see — the building you're in, the furniture around you, the devices you use — was designed by an engineer. Someone thought about the problem, sketched solutions, chose materials, built a prototype, tested it, and improved it.

You've now done exactly that.

PART 4 - MATH, CODING AND LOGIC

Here is something nobody tells you often enough: mathematics is one of the most creative things humans do.

We often teach math as a set of procedures: follow these steps, get this answer. But this misses what mathematics actually is — a language for describing patterns, structures, and relationships. Math is how we understand the shape of a galaxy, the spread of a disease, the fairness of an election, the structure of a piece of music, and the security of every message you send on your phone.

Coding is a close cousin: it is the act of giving precise instructions to a machine, and in doing so, creating something that didn't exist before. Every app on your phone, every website you visit, every game you play began as an idea in someone's mind that they then translated, carefully, into a language a computer could follow.

The ten ideas in this part are about learning to see the world through the lens of patterns and logic. They include ciphers and encryption (how messages are hidden and revealed), computer animation and game design, symmetry and fractals, probability, and data visualization.

You don't need to be a math prodigy for any of this. You need curiosity, patience, and willingness to think systematically. These are skills, not talents — they get stronger the more you use them.

IDEA 31 — THE CAESAR CIPHER: YOUR FIRST ENCRYPTION

Where: Desk, with paper and pencil — no computer needed

Time: 30–45 minutes

Mess level: None

The Story:

Julius Caesar used a simple encryption method to send secret military messages over 2,000 years ago. The idea: shift every letter in the message forward by a fixed number in the alphabet. To decode it, you shift backward by the same amount.

This is called a *Caesar cipher*, and while it is too simple to use for real secrets today, understanding it gives you the foundation for understanding all of modern cryptography.

How It Works:

1. **Choosing a "key".** Pick a number. This is your key — how many letters you shift. Let's use ****3****.

2. **Encryption.**

A → D

B → E

C → F

...

X → A (wrap around!)

Y → B

Z → C

So the message ****HELLO**** becomes ****KHOOR****.

3. **Decryption:** Shift each letter back by 3.

K → H, H → E, O → L, O → L, R → O

****KHOOR**** → ****HELLO****

Make a Cipher Wheel:

1. Cut two circles from cardboard — a larger outer circle and a smaller inner one.
2. Write A–Z around the edge of the outer circle (equally spaced).
3. Write A–Z around the inner circle in the same way.
4. Connect them with a paper fastener through the center so the inner circle rotates.
5. Set the inner circle so that A on the inner circle aligns with a letter on the outer circle — that's your key.

Now you can encrypt messages instantly using your wheel.

Activities:

1. Encrypt your full name using key = 5.
2. Write a secret message for a friend. Give them the key number separately. Can they decode it?
3. Can you decode this message without knowing the key? (This is called *codebreaking*.) The message: ****WKXQGHUVWRUP**** (key = 3). Hint: try all 25 possible shifts.
4. How many possible keys does a Caesar cipher have? Why does this make it weak?

Going Deeper: Why Simple Ciphers Fail

A Caesar cipher is vulnerable to **frequency analysis** — the observation that in any language, certain letters appear much more often than others. In English, E, T, A, O, I, N are most common. In a long encrypted message, the most common cipher letter probably represents E. This technique alone can crack a Caesar cipher in minutes, even without knowing the key.

Modern encryption (like the RSA encryption protecting your bank transactions and messages) uses mathematical operations involving very large prime numbers — operations that are easy to do but practically impossible to reverse without the key. The gap between easy and hard computation is what makes modern cryptography secure.

Ada Lovelace's vision of using mathematics for computation laid the philosophical foundation for all of this.

What You Learned: Cryptography, substitution ciphers, encryption and decryption, modular arithmetic, frequency analysis, the basis of modern cybersecurity.

IDEA 32 — LOGIC GRID PUZZLES

Where: Desk, with paper or printed grid

Time: 20–45 minutes per puzzle

Mess level: None

What Is a Logic Grid Puzzle?

A logic grid puzzle gives you a set of clues and asks you to figure out how different items are matched — which person has which pet, which job, lives in which house, and so on. No guessing is needed: pure logical deduction takes you to the answer.

Example Puzzle (Beginner): Three girls — Anh, Binh, and Chi — each won a prize in a different subject: Math, Science, and Literature.

Clues:

1. Anh did not win in Science.
2. The Science winner is not Binh.
3. Chi won the Literature prize.

Use a grid:

	Math	Science	Literature
Anh			
Binh			
Chi			

Working through it:

From clue 3: Chi won Literature.

	Math	Science	Literature
Anh			
Binh			
Chi	✗	✗	✓

From clue 2: Binh did not win Science. Mark Binh/Science = ✗.

	Math	Science	Literature
Anh			
Binh		✗	
Chi	✗	✗	✓

Since neither Chi nor Binh won Science, Anh must have won Science. Mark Anh/Science = ✓. Mark Anh/Math = ✗, Anh/Literature = ✗.

	Math	Science	Literature
Anh	✗	✓	✗
Binh		✗	

Chi	✗	✗	✓
-----	---	---	---

Since Binh didn't win Science or Literature, Binh won Math.

	Math	Science	Literature
Anh	✗	✓	✗
Binh	✓	✗	✗
Chi	✗	✗	✓

Answer: Anh = Science, Binh = Math, Chi = Literature.

Why Logic Puzzles Matter: Logic puzzles develop **deductive reasoning** — the ability to derive conclusions with certainty from premises. This is:

- The foundation of mathematical proof
- Essential in programming (debugging requires methodical elimination of possibilities)
- Useful in everyday decision-making
- The core of legal argumentation and scientific reasoning

The process of marking what's impossible and narrowing down to what must be true is called **proof by elimination**, and it is one of the most powerful tools in mathematics.

Your Challenges: Four girls — Dung, Lan, Mai, Phuong — each have a different after-school activity: coding, painting, basketball, and chess. They each come from a different grade: 5, 6, 7, 8.

Clues:

1. The grade 7 student plays basketball.
2. Lan is in grade 5.
3. Mai does coding.
4. The chess player is older than the basketball player.
5. Phuong is in grade 8
6. Dung does not paint

(Work it out! The full solution is at the end of this chapter.)

What You Learned: Deductive reasoning, proof by elimination, logical thinking, grid-based problem solving, the foundations of formal logic.

IDEA 33 — SIMPLE ANIMATION WITH SCRATCH

Where: Computer with internet access (or tablet/phone for Scratch app)

Time: 60–90 minutes

Mess level: None

What Is Scratch?

Scratch is a free, visual programming language developed by MIT (Massachusetts Institute of Technology) that lets you create animations, games, and interactive stories by snapping together code blocks — like digital LEGO. It is used by millions of people around the world, including professional developers who use it to prototype ideas.

Visit: scratch.mit.edu (free, no download required)

Project: A Walking Character Animation

This project teaches you the core concepts of animation: *frames, loops, and timing.*

Step 1: Choose Your Sprite

In Scratch, animated characters are called *sprites*. Click "Choose a Sprite" and pick a character (the cat that comes by default works perfectly). You can also draw your own sprite.

Step 2: Make It Walk

Many sprites have multiple **costumes** — different versions of the image at different positions (this is what creates the illusion of movement in animation).

Click on the "Costumes" tab to see your sprite's costumes. If it has walking costumes (like the default cat, which has "costume1" and "costume2"), you can animate it.

Step 3: Write the Code

Click the "Code" tab and build this program using snap-together blocks:


```
when [green flag clicked]
forever
  switch costume to [costume1]
  wait [0.2] seconds
  switch costume to [costume2]
  wait [0.2] seconds
end
```

Click the green flag. Your character should now appear to walk in place — switching between two poses.

Step 4: Add Movement

Now add movement so it walks across the screen:

```
when [green flag clicked]
set x to [-200]
forever
  change x by [3]
  switch costume to [costume1]
  wait [0.1] seconds
  change x by [3]
  switch costume to [costume2]
  wait [0.1] seconds
  if <(x position) > [200]> then
    set x to [-200]
  end
end
end
```

This makes your character walk from left to right and loop back.

Understanding the Code:

- `when [green flag clicked]` — this is an **event trigger**. The code begins when you click the flag.
- `forever` — this is a **loop**. Everything inside runs again and again, forever, until you stop.
- `wait [0.2] seconds` — **timing control** that determines animation speed.

- `if...then` — a **conditional statement**. It checks a condition and only runs the code inside if that condition is true. This is one of the most fundamental structures in all programming.
- `change x by [3]` — **variables and coordinates**. X position tells the sprite where it is horizontally.

What to Try Next:

- Change the wait time. How does animation speed change?
- Add a background from the library.
- Make the character stop when it touches the edge.
- Add sound when the character moves.

What You Learned: Loops, conditionals, event triggers, variables, animation principles, computational thinking.

IDEA 34 — MAKE YOUR FIRST GAME IN SCRATCH

Where: Computer

Time: 90–120 minutes

Mess level: None

Project: Catch the Star

A ball (the player) moves with arrow keys. Stars fall from the top of the screen. The player catches stars to score points. Stars that fall off the bottom of the screen are lost.

What You'll Build:

The Player Sprite:

```
when [green flag clicked]
  forever
    if <key [left arrow] pressed?> then
      change x by [-5]
    end
    if <key [right arrow] pressed?> then
      change x by [5]
    end
  end
end
```

The Star Sprite:

```
when [green flag clicked]
  set y to [180]
  set x to [pick random (-220) to (220)]
  forever
    change y by [-3]
    if <touching [player]> then
      change [score] by [1]
      set y to [180]
      set x to [pick random (-220) to (220)]
    end
  end
```

```
if <(y position) < [-180]> then
  set y to [180]
  set x to [pick random (-220) to (220)]
end
end
```

The Score Display:

- Create a variable called "score" (in the Variables blocks menu).
- The score variable will appear on screen and update automatically.

Game Design Concepts:

- **Game loop:** Most video games run a continuous loop: check input → update game state → draw screen → repeat. Your Scratch 'forever' loops do exactly this.
- **State variables:** The score, positions of sprites, whether the game is won or lost — these are all **states**. Variables store state.
- **Collision detection:** Checking touching [player]? is collision detection — one of the most important (and complex to get right) problems in game development.
- **Randomness:** Using pick random makes gameplay unpredictable and therefore interesting. Games use random numbers constantly.

Extend Your Game:

- Add increasing difficulty: make the stars fall faster as score increases.
- Add a timer: the game ends after 30 seconds.
- Add different types of stars worth different points.
- Add sound effects.
- Design your own sprites instead of using default ones.

What You Learned: Game loops, variables, collision detection, randomness, user input, game design, object-oriented thinking.

IDEA 35 — SYMMETRY IN ART AND MATHEMATICS

Where: Desk or art table

Time: 45–60 minutes

Mess level: Low (paints possible)

Mathematical Symmetry:

A shape has **symmetry** if you can perform a transformation on it that leaves it looking exactly the same. The main types:

- **Line symmetry (reflective symmetry):** You can draw a line through the shape and each half is a mirror of the other. A butterfly has line symmetry. A human face nearly does (but not quite).
- **Rotational symmetry:** You can rotate the shape around a central point and it looks the same before completing a full rotation. A starfish has 5-fold rotational symmetry — it looks the same after rotating 72° , 144° , 216° , or 288° .
- **Translational symmetry:** You can slide a pattern along a surface and it looks the same. Wallpaper patterns and tile floors use this.
- **Point symmetry:** Rotating 180° around a central point gives the same shape.

Activity 1: Symmetry Hunt

Look around your home and find:

- 3 objects with line symmetry
- 2 objects with rotational symmetry
- 1 example of translational symmetry

Look for symmetry in nature: flowers (how many fold symmetry?), leaves, snowflakes (6-fold symmetry — always!), shells.

Look in art: Islamic geometric art, traditional embroidery patterns, mandalas — all built on mathematical symmetry principles.

Activity 2: Create a Symmetrical Artwork

Fold and paint:

1. Fold a piece of paper in half.
2. Drop blobs of paint on one half.
3. Fold, press firmly, unfold.
4. You get a perfectly symmetrically mirrored image — this is line symmetry through the fold line.

Rotational mandala:

1. Using a compass, draw circles of different sizes centered at one point.
2. Divide the circle into equal sections (6 sections = 60° , 8 sections = 45° , 12 sections = 30°).
3. Design one section, then repeat identically in every section.
4. Color using a symmetrical color scheme.

Symmetry in Science: Symmetry isn't just beautiful — it is scientifically fundamental:

- **Biology:** Most animals have bilateral symmetry (left-right mirror images). This is thought to reflect developmental efficiency and may be associated with fitness.
- **Chemistry:** The symmetry of a molecule determines its properties — including its smell, taste, and how it interacts with other molecules. Two molecules that are mirror images of each other (called *enantiomers*) can have completely different biological effects.
- **Physics:** The deepest laws of physics are expressions of symmetry. Noether's Theorem (proved by mathematician Emmy Noether in 1915) shows that every fundamental conservation law (conservation of energy, momentum, etc.) corresponds to a symmetry in the laws of physics.

What You Learned: Types of symmetry, symmetry in nature and art, mathematical properties, applications in science.

IDEA 36 — FRACTALS: THE INFINITY IN SIMPLE RULES

Where: Paper and pencil, OR a computer (optional)

Time: 30–60 minutes

Mess level: None

What Is a Fractal?

A fractal is a pattern that looks similar at every scale — if you zoom in, you see the same pattern repeating. Fractals are created by applying a simple rule over and over again (in math, this is called **iteration**).

Fractals appear everywhere in nature: coastlines, mountain ranges, clouds, lightning bolts, snowflakes, river networks, the branching of trees and blood vessels and lung airways, the patterns in broccoli (look at a Romanesco broccoli closely!).

Draw a Koch Snowflake:

The rule: Take a line. Replace the middle third with two sides of an equilateral triangle.

1. Draw a straight line.
2. Divide it into thirds. Remove the middle third. In its place, draw two sides of an equilateral triangle (making a bump in the middle). You now have 4 line segments where you had 1.
3. Apply the same rule to every line segment. You now have 16 line segments.
4. Apply again to every segment. 64 line segments.

Start with an equilateral triangle and apply this rule to every side. After several iterations, you get the Koch Snowflake — a shape with infinite perimeter but finite area.

(Think about that: infinite perimeter, finite area. This is counterintuitive but mathematically provable. This is the kind of thing that makes mathematics strange and wonderful.)

Draw a Sierpinski Triangle:

The rule: Take a filled triangle. Remove the middle triangle (the upside-down triangle made by connecting the midpoints of all three sides). Apply the same rule to the three remaining triangles. Repeat.

1. Filled triangle.
2. 3 filled triangles + 1 empty middle.
3. 9 filled triangles.
4. 27 filled triangles.

After infinite iterations, the total area of filled triangles approaches zero — but the overall triangle shape remains!

Try on Computer:

Search for "fractal explorer online" to see fractals like the Mandelbrot Set — the most famous fractal in mathematics, where a tiny, simple mathematical rule creates infinite complexity.

You can zoom into any edge of the Mandelbrot Set forever, always finding new patterns.

What You Learned: Fractals, iteration, self-similarity, infinite perimeter/finite area, fractals in nature, mathematics of complex systems.

IDEA 37 — PROBABILITY WITH COINS AND DICE

Where: Desk or table

Time: 45–60 minutes

Mess level: None

The Key Idea:

Probability measures how likely something is to happen. It ranges from 0 (impossible) to 1 (certain). A probability of 0.5 means equally likely to happen or not happen.

The probability of an event = (number of favorable outcomes) ÷ (total number of possible outcomes)

Example: Rolling a 3 on a standard die: 1 favorable outcome ÷ 6 possible outcomes = $1/6 \approx 0.167$

Experiment A: Coin Flip

Prediction: If you flip a coin 100 times, how many heads will you get? Most people say "50" — but will you actually get exactly 50?

Flip a coin 100 times. Tally heads and tails. Record the result after every 10 flips.

Flips	Heads so far	Tails so far	% Heads
10			
20			
30			
40			
50			
60			
70			

80			
90			
100			

What do you notice? After 10 flips, your result may be far from 50%. After 100, it should be closer. This illustrates the **Law of Large Numbers** — as sample size increases, the observed frequency approaches the theoretical probability.

Experiment B: Dice and Expected Value

Roll two dice 50 times. Record the sum each time. Which sum appears most often? Why?

There are 36 possible outcomes when rolling two dice (6×6). The sum 7 can be made in 6 ways (1+6, 2+5, 3+4, 4+3, 5+2, 6+1). The sum 2 can only be made in 1 way (1+1). So the probability of rolling a 7 is $6/36 = 1/6$, while rolling a 2 is $1/36$.

Draw a bar chart of your results. Does it match the theoretical prediction?

This is how casinos are designed: using probability, they know that over thousands of games, they will profit — the "house edge." Over a single game, anything can happen. Over many games, probability always wins.

Experiment C: Random or Not?

Write down a sequence of 30 coin flips from your imagination (H, T, H, H, T...) without actually flipping. Then write down 30 actual coin flips.

Now compare: in a truly random sequence, you will often see runs of 5, 6, or 7 of the same result in a row. In an imagined "*random*" sequence, people usually avoid long runs — because our brains perceive them as "*not random*." But they are perfectly consistent with randomness.

This is why humans are very bad at intuiting probability — our brains look for patterns even in randomness.

What You Learned: Probability, Law of Large Numbers, expected value, experimental vs theoretical probability, randomness and human perception.

IDEA 38 — COLLECTING AND ANALYZING CLASS DATA

Where: School or home (with family)

Time: 45–60 minutes

Mess level: None

The Project: Become a statistician. Choose a question, collect data from your classmates or family, analyze it, and present your findings.

Choose a Research Question:

Some ideas:

- How many hours of sleep do students in your class get each night? Does it vary by grade?
- What is the most popular after-school activity in your class?
- How do students travel to school? What is the most common method?
- What is the range of heights in your class?
- How many languages can students in your class speak?

Collect Your Data:

Create a simple survey. Collect responses from at least 15–20 people.

Tip on survey design: Be specific.

"Do you sleep enough?" is vague.

"How many hours of sleep did you get last night?" gives measurable data.

Analyze Your Data:

Calculate:

- **Mean:** Add all values, divide by number of values. (Average.)
- **Median:** Arrange values in order. The middle value. (Half the values are above, half below.)
- **Mode:** The most commonly occurring value.

- **Range:** Highest value minus lowest value.

Which measure is most useful for your question? They each tell you something different. For example, if 19 students sleep 7–8 hours and 1 student sleeps 2 hours, the mean will be pulled down by that outlier — but the median will better represent the typical student.

Display Your Data:

- For **categories** (transport types, favorite subjects): bar chart or pie chart
- For **continuous measurements** (height, sleep hours): histogram or box plot
- For showing **individual data points**: dot plot

Draw Conclusions:

- Write a short summary:
- What did you find?
- Were there any surprises?
- What patterns emerged?
- What further questions does this raise?
- What limitations does your data have? (Small sample? Self-reported? Possible bias?)

What You Learned: Data collection, survey design, descriptive statistics, mean/median/mode/range, data visualization, drawing conclusions from data.

IDEA 39 — ANALYZING YOUR OWN PHONE / SCREEN TIME DATA

Where: Home

Time: 30–40 minutes

Mess level: None

The Idea: Your smartphone already collects data about you. In Settings (on most smartphones), you can find **Screen Time** (iPhone) or **Digital Wellbeing** (Android) data showing:

- Total daily screen time
- Breakdown by app category
- How many times you pick up your phone each day
- Which apps you use most

This is real data about your own behavior. Analyzing it is a genuine data science exercise.

Collect Data Over One Week:

- Every day for one week, record (or screenshot) your screen time data:
- Total screen time (hours and minutes)
- Time on social media
- Time on entertainment/video
- Time on games
- Time on education/productivity
- Number of pickups

Analyze:

1. Calculate your daily average total screen time.
2. What percentage is each category?
3. Is there a day of the week pattern? (More screen time on weekends?)
4. Does your screen time correlate with how productive or happy you felt that day?

Reflect:

- Does your screen time data match your perception of how much you use your phone?
(Most people underestimate their screen time significantly — this is a well-documented cognitive bias.)
- How does your data compare to published statistics on average screen time for your age group?
- What changes, if any, do you want to make based on what you found?

The Bigger Picture:

Your phone data is just one example of what is now called **personal data** — the massive amount of information digital devices collect about us. This data is used by companies to serve ads, by researchers to study behavior, and by governments for various purposes. Understanding data — how it's collected, what it means, and what its limitations are — is a critical skill for life in the 21st century.

What You Learned: Data collection, personal analytics, screen time patterns, data literacy, cognitive bias, critical thinking about technology.

IDEA 40 — MAKING DATA VISIBLE: BUILDING A DATA VISUALIZATION

Where: Computer (spreadsheet software) or paper and colored pens

Time: 60–90 minutes

Mess level: None

The Power of Visualization:

Numbers on a page are hard to understand quickly. A well-designed chart or graph communicates the same information instantly — showing patterns, comparisons, and stories that are invisible in the raw numbers.

Data visualization is now one of the most important skills in science, business, journalism, and public health. The scientists and analysts who communicated COVID-19 data to the public were doing data visualization.

Project: Create an Infographic Poster

Choose a dataset that matters to you. Some ideas:

- Climate data (average temperatures in your city over 50 years)
- Population data (your country's population by age group)
- Gender gaps in STEM fields (% of women in different careers)
- Your class's data from Idea 38

Choosing the Right Chart:

What you want to show	Best chart type
Comparison between categories	Bar chart
Change over time	Line chart
Part of a whole	Pie chart
Relationship between two variables	Scatter plot
Distribution of values	Histogram

Design Principles:

Accuracy: Your chart must accurately represent the data. Truncating the y-axis (starting at a number other than 0) can make small differences look dramatic — this is a common way charts mislead.

Clarity: Remove everything that doesn't help the reader understand the data. Label axes clearly. Use a legend if needed.

Honesty: Don't choose colors, scales, or chart types that exaggerate your findings or lead the reader to incorrect conclusions.

Story: The best data visualizations tell a clear story. What is the one thing you want your reader to take away?

Using Spreadsheet Software:

In Google Sheets or Excel:

1. Enter your data in rows and columns.
2. Select the data.
3. Insert → Chart.
4. Choose chart type.
5. Customize colors, labels, and title.

Create a Paper Version:

If you don't have access to software, create your visualization by hand on poster paper:

- Use graph paper as a guide for bar charts.
- Use a compass for pie charts.
- Use rulers for line charts.
- Add color for clarity and visual appeal.

What You Learned: Data visualization principles, chart types and their uses, misleading charts, storytelling with data, spreadsheet tools.

End of Part 4 — Math, Coding and Logic

Mathematics is not just about getting the right answer. It is about seeing the patterns in the world around you, and having tools to describe those patterns precisely.

Coding is not just about making apps — it is about learning to break any complex problem into small, logical steps.

Whether you become a data scientist, a game developer, a biologist, an economist, or something entirely different — the thinking skills you've practiced in this part will serve you for the rest of your life.

IDEA 32 Challenge answer:

Dung - 7 - basketball

Lan - 5 - paint

Mai - 6 - coding

Phuong - 8 - chess

PART 5 - STEM FOR A BETTER WORLD

We've saved this part for last, because it is in some ways the most important.

All the science, engineering, mathematics, and technology in the previous four parts are tools. What matters is what we do with those tools.

The world you are growing up in faces extraordinary challenges: climate change, plastic pollution, biodiversity loss, water scarcity, food insecurity, and persistent inequality in access to education and resources. These are not small problems. They are urgent, complex, and interconnected. They will not be solved by one country, one organization, or one generation.

They will be solved by people — people with scientific knowledge, engineering creativity, mathematical precision, and most importantly, the will to use their skills for the benefit of all.

Girls and women have a critical role to play in this work. Research on environmental leadership and community-based science consistently shows that women are often among the most effective advocates for environmental protection and sustainable development — and are also among the most affected by environmental degradation. Bringing women fully into the conversation and the leadership of these challenges is not just a matter of fairness. It is a matter of effectiveness.

The ten ideas in Part 5 invite you to do STEM on purpose — with a specific intention to understand and improve the world around you. Some projects are small. Some might grow into something larger. All of them connect you to some of the most important scientific and engineering work happening in the world right now.

You are not too young to contribute. The most important environmental activists of the 21st century include many who started their work as teenagers — and the most

impactful scientists solving environmental problems are doing work that builds, step by step, on exactly the kind of curiosity and observation you'll practice here.

IDEA 41 — MAKING ECO-ENZYME

Where: Home (needs 3 months of patience!)

Time: Setup: 30 minutes. Waiting: 3 months. Worth it!

Mess level: Low

What Is Eco-Enzyme?

Eco-enzyme (also called garbage enzyme or fruit enzyme) is a liquid produced by fermenting fruit and vegetable scraps with sugar and water. The result is a multipurpose natural cleaner, fertilizer, and pest deterrent that uses materials you would otherwise throw away.

This technique was developed and popularized by Dr. Rosukon Poompanvong in Thailand and is now used by environmental communities across Southeast Asia, including Vietnam, Malaysia, and Indonesia.

What You Need:

- 1 part brown sugar or molasses (e.g., 100g)
- 3 parts fresh fruit/vegetable scraps (e.g., 300g) — citrus peels work exceptionally well; avoid cooked food, meat, or oily scraps
- 10 parts water (e.g., 1000ml = 1 liter)
- A plastic bottle or jar with a loose-fitting lid (not airtight — it needs to breathe)
- A label and marker

What To Do:

1. Dissolve the sugar in the water in your container.
2. Add the fruit and vegetable scraps.
3. Close loosely. Label with the date and contents.
4. For the first month, open the container daily to stir and release built-up gas.
5. After 3 months, strain out the solid material. The liquid is your eco-enzyme.

The solids can be composted.

What's Happening?

The sugar feeds **fermentation** — a process in which microorganisms (mostly beneficial bacteria and yeast) break down the organic material and produce:

- **Acids** (which give eco-enzyme its cleaning properties)
- **Enzymes** (which break down organic stains and odors)
- **Alcohol** (in small amounts, as a byproduct)

The citrus peels add limonene — a natural solvent that cuts grease.

Uses for Eco-Enzyme:

- **Floor cleaner:** Dilute 1 part eco-enzyme with 10 parts water.
- **Plant fertilizer:** Dilute 1 part eco-enzyme with 500 parts water and apply to plants.
- **Fruit and vegetable wash:** Dilute in water to remove pesticide residue.
- **Drain cleaner:** Pour undiluted down drains to clear buildup.
- **Pest deterrent:** Spray diluted on plants.

The Bigger Picture: Organic waste (food scraps) sent to landfill decomposes anaerobically (without oxygen), producing methane — a greenhouse gas around 25 times more potent than CO₂ over a 100-year period. Eco-enzyme production is one of hundreds of ways individuals and communities can reduce organic waste while creating something useful.

Keep a production log: Record the date you started, the ingredients you used, observations during fermentation (color changes, smell, gas production), and your assessment of the final product.

What You Learned: Fermentation, microbiology, organic chemistry, waste reduction, natural cleaning products, circular economy principles.

IDEA 42 — PLASTIC WASTE SURVEY IN YOUR SCHOOL OR NEIGHBORHOOD

Where: School, home neighborhood, park, or market

Time: 2 hours for data collection; 1–2 hours for analysis

Mess level: None (bring gloves!)

What Is Citizen Science?

Citizen science is scientific research conducted by members of the public, often in collaboration with or under the guidance of professional researchers. It allows data to be collected at a scale impossible for small research teams — and it gives ordinary people the power to contribute to real scientific knowledge.

Your plastic waste survey is a genuine citizen science project.

Design Your Survey:

Research question:

- What is the composition of plastic waste in [*your location*]?
- Are certain types more common?
- Where does it concentrate?

Survey area: Choose a defined area — your school yard, a section of your street, a park, a market.

Categories to record: For each piece of plastic waste you observe and/or collect, record:

- Type of plastic (bottle, bag, food packaging, straw, cup, other)
- Estimated size (small <10cm, medium, large)
- Condition (intact, broken, very degraded)
- Location found (ground, drainage, near bin, etc.)
- Brand if visible (optional — this can be used for brand accountability data)

Sample area and time: Be consistent. Survey the same area at the same time each week, or survey different areas using the same time limit.

Collect Your Data: Take gloves. Take a bag (ideally for recycling/sorting). Take your notebook.

Walk the survey area carefully. Count and record each piece of plastic waste. If safe to do so, collect it for closer categorization.

Analyze:

- What is the total number of pieces?
- What are the most common types?
- Is there a pattern in where plastic concentrates?
- How does it compare week to week?

Present and Share: Create a presentation or poster of your findings to share with your class, school, or community. Data presented clearly can be a powerful tool for change.

Consider: Could you share your data with your local government, school administration, or an environmental organization?

Connect to the Bigger Picture: An estimated 8 million metric tons of plastic enter the ocean every year. Microplastics — tiny fragments of degraded plastic — have now been found on the highest peaks of the Himalaya, in the deepest trenches of the ocean, in the air of remote Arctic wilderness, and in human blood. This is one of the most widespread pollution challenges in history.

Your survey contributes to understanding the local source of this global problem.

What You Learned: Citizen science methodology, data collection design, environmental surveying, waste composition analysis, plastic pollution awareness, scientific communication.

IDEA 43 — TESTING INSULATING MATERIALS

Where: Home or school

Time: 60–90 minutes

Mess level: Low

The Question: Which everyday material is the best thermal insulator?

Why It Matters: Thermal insulation is fundamental to energy efficiency in buildings. Poor insulation means energy (and money) is constantly lost — heating systems work harder to maintain temperature. In developing countries, poor building insulation contributes significantly to energy poverty. In the context of climate change, improving building insulation is one of the most cost-effective ways to reduce energy use and carbon emissions.

What You Need:

- Several identical small cans or cups (yogurt containers work)
- Hot water (the same temperature each time — boil and let cool to exactly the same temperature)
- A thermometer
- Insulating materials to test: wool fabric, cotton fabric, aluminum foil, newspaper (crumpled), bubble wrap, polystyrene (styrofoam), plastic bag, nothing (control)
- A timer
- A notebook

What To Do:

1. Wrap identical containers in each insulating material. The control is unwrapped.
2. Pour hot water at the same temperature into each container.
3. Record the starting temperature.
4. Measure and record temperature every 5 minutes for 30 minutes.

5. The container that loses heat most slowly has the best insulation.

Record Your Results:

Time (min)	Control	Wool	Foil	News- paper	Bubble wrap	Styro- foam
0						
5						
10						
15						
20						
25						
30						

Plot all results on a single graph (temperature on y-axis, time on x-axis, one line per material). The line that drops least steeply is the best insulator.

What's Happening?

Heat moves by three mechanisms:

- **Conduction:** Direct transfer through contact. Materials with tightly packed molecules conduct well (metals). Materials with trapped air pockets conduct poorly.
- **Convection:** Heat transfer through moving fluid (air or liquid). Good insulators prevent air from circulating.
- **Radiation:** Emission of infrared radiation. Foil reflects radiation.

The best insulators (wool, styrofoam, aerogel in advanced applications) work by trapping small pockets of air — because air is an excellent insulator. The material itself isn't always what's insulating; the trapped air is.

What You Learned: Thermal insulation, heat transfer mechanisms, controlled experiment design, data graphing, materials science, energy efficiency.

IDEA 44 — BUILDING A MODEL GREEN HOUSE

Where: Home, with access to recycled materials

Time: 2–3 hours

Mess level: Medium

The Challenge: Design and build a model house (at 1:50 scale, so 1 cm in the model = 50 cm in reality) that incorporates as many sustainable design features as possible:

- Solar orientation (windows facing south in the northern hemisphere to maximize winter sun)
- Natural ventilation (openings placed to create cross-ventilation)
- Thermal mass (heavy materials inside that absorb heat during the day and release it at night)
- Green roof (a layer of plants on the roof for insulation and stormwater management)
- Rainwater collection (a sloped roof that channels water to a tank)
- Natural light optimization (skylights, light-colored interior surfaces)

Materials:

- | | |
|----------------------|---------------------|
| ● Recycled cardboard | ● Sand |
| ● Foam | ● Soil |
| ● Fabric scraps | ● Small plants/moss |
| ● Plastic packaging | ● Bottle caps |
| ● Clay | ● Foil |
| | ● Fabric |

Design First: Sketch a floor plan and elevation (front view) before building. Label every sustainable feature and explain why you included it.

Consider:

- Which direction does sunlight come from at your location?
- How do breezes typically flow in your area?

- What local materials and techniques are traditionally used in your region?
(Traditional Vietnamese architecture, for example, incorporates many passive cooling principles.)

Build and Evaluate:

- After building, use a lamp to simulate sunlight:
- Test whether your sunlight-facing windows actually let more light in.
- Test whether your ventilation openings create airflow (you can use a lit incense stick and watch the smoke — with adult supervision).
- Measure interior temperature versus an uninsulated control box.

What You Learned: Sustainable architecture, passive design, solar orientation, thermal mass, green building principles, energy efficiency, scale models.

IDEA 45 — EXPLORING SOLAR ENERGY WITH A MINI MODEL

Where: Home, outdoors in sunlight

Time: 60–90 minutes

Mess level: Low

Project A: Solar Thermal — Heating Water with Sunlight

Build a simple solar water heater using a coil of black hose or tubing:

1. Paint or wrap a length of plastic tubing in black (or use pre-blackened tubing).
2. Coil it on a dark-colored backing board inside a transparent plastic bag or covered in clear plastic wrap (to create a greenhouse effect).
3. Run water through the coil using gravity (position input higher than output, or use a small aquarium pump).
4. Measure the temperature of water entering and leaving the coil in full sunlight.

This is the same principle used in **solar thermal panels** on rooftops around the world to heat water for homes.

Project B: Solar Photovoltaic — Powering a Small Device

If you can access a small solar panel (available cheaply from electronics suppliers or salvaged from solar garden lights):

1. Connect the solar panel to a small LED or motor.
2. Measure the output in full sun, partial shade, and complete shade.
3. Try different angles of the panel relative to the sun. At what angle is output maximum?

The angle that gives maximum output is the angle at which the panel faces the sun most directly — **perpendicular to the incoming rays**. This is why large solar installations track the sun's movement across the sky.

The Bigger Picture: Solar energy is now the cheapest source of electricity in history — cheaper than coal in most countries. The dramatic cost reduction (solar panel

prices fell by more than 90% between 2010 and 2023) is one of the most remarkable technological developments in history, driven by improvements in manufacturing, materials science, and economies of scale.

Vietnam, as of recent years, has become one of the fastest-growing solar energy markets in the world — solar power now represents a significant portion of the country's electricity generation.

What You Learned: Solar energy principles, solar thermal vs photovoltaic, panel angle and efficiency, energy conversion, renewable energy technology.

IDEA 46 — SIMPLE WATER QUALITY TESTING

Where: Home or school; water samples from different sources

Time: 60–90 minutes

Mess level: Low (avoid getting test chemicals on skin)

Testing Parameters: You will test water samples from different sources (tap water, rain water, river/pond water if available, bottled water) for:

1. **pH** — using your red cabbage indicator or pH strips
2. **Turbidity** — how cloudy/clear the water is (visual assessment or a simple light test)
3. **Dissolved solids (TDS)** — if you have a basic TDS meter (inexpensive, available online)
4. **Presence of contaminants** — iron test with specific test strips available from hardware stores; chlorine test with pool test kits

Simple Turbidity Test:

1. Print or draw a small symbol (a star or letter) on white paper.
2. Place the paper under your water sample in a clear glass.
3. Look down through the water from the top.
4. Record: Can you see the symbol clearly, somewhat, barely, or not at all?
5. This gives you a relative turbidity measure.

Create a Water Quality Report: For each water source, record all your measurements. Create a "Water Quality Report" in table form, like an official scientific document. Include:

- Source name and collection location
- Date and time of collection
- Results of each test
- Comparison to WHO or national drinking water standards (look these up)
- Your conclusions: Is this water likely safe? What treatment would be needed?

What You Learned: Water quality parameters, pH, turbidity, TDS, water testing methods, comparison to safety standards, environmental analysis.

IDEA 47 — GROWING PLANTS WITHOUT SOIL: MINI HYDROPONICS

Where: Home, indoors

Time: Setup: 1 hour. Growing period: 2–4 weeks

Mess level: Low (water careful)

What Is Hydroponics?

Hydroponics is the practice of growing plants in a nutrient-rich water solution rather than soil. The plant's roots are submerged in or regularly supplied with water containing all the minerals the plant needs. Without soil, plants can grow faster, use much less water, and be grown anywhere — including indoors, in cities, in deserts, or in space.

Hydroponic farming is one of the most promising technologies for sustainable food production in the 21st century. It uses 90% less water than conventional farming, can be done in urban areas (eliminating transportation emissions), and can produce crops year-round regardless of climate.

What You Need:

Deep water culture system (simplest setup):

- A dark-colored plastic container (to block light and prevent algae)
- Net pots or cups with holes (or make holes in styrofoam to hold cups)
- A small aquarium air pump with tubing and air stone (keeps water oxygenated)
- Hydroponic nutrient solution (available online or at garden shops — or you can research and mix your own mineral salts)
- Growing medium: rockwool, clay pebbles, or clean gravel
- Seeds: lettuce, spinach, cress, or herbs work best for beginners
- Water

Setup:

1. Fill the container with nutrient solution.

2. Fit your net pots into the lid so the bottom of each pot is just touching the solution.
3. Place a small piece of rockwool or gravel in each pot. Place 2–3 seeds on the growing medium.
4. Run the air pump to oxygenate the water.
5. Cover the container (light on the plants, dark on the water).
6. Within 3–7 days, seeds should sprout. Within 2–4 weeks, you may be able to harvest!

Experiment Within the Project:

- Grow the same plant hydroponically and in soil. Compare growth rate, leaf size, and yield.
- Try different nutrient concentrations. Too little: slow growth. Too much: nutrient burn.
- Try different light sources: natural sunlight vs. LED grow light.

Record Keeping: Every 2 days, record:

- Plant height
- Number of leaves
- Color of leaves (pale yellow = nutrient deficiency)
- Root length (visible through clear growing medium)
- Water level
- Color

What You Learned: Plant nutrition, hydroponics technology, photosynthesis, sustainable agriculture, plant growth factors, experimental comparison.

IDEA 48 — CREATING USEFUL PRODUCTS FROM RECYCLED MATERIALS

Where: Home

Time: 1–2 hours

Mess level: Medium

The Circular Economy:

The traditional economic model is linear:

make → use → throw away

In a circular economy, materials cycle:

make → use → recover → remake

Upcycling — transforming waste materials into something of higher value — is a core principle of the circular economy.

Your Challenge: Design and make one useful, beautiful, or innovative product using only materials that would otherwise be discarded. The product should have genuine use — it should solve a real problem or meet a real need.

Ideas and Inspiration:

From plastic bottles:

- A vertical garden planter (cut bottles, mounted on a wall with soil and plants)
- A desk organizer (cut, sand smooth, decorate)
- A bird feeder
- A self-watering plant pot (bottom half holds water, top half holds soil with a wick)

From newspaper/cardboard:

- Strong storage boxes (origami techniques for cardboard)
- Seedling starter pots (biodegradable — plant directly in soil)
- Decorative paper beads (tightly rolled paper coated in varnish)

From glass jars:

- Solar-charged fairy light lanterns
- Herb garden
- Desk organizer

From fabric scraps:

- Tote bags (zero-sew versions using fabric glue)
- Beeswax wraps (as a sustainable alternative to plastic wrap)
- Small pouches for carrying seeds, pens, or jewelry

Document Your Process:

Take photos at each stage. Write or draw:

1. The problem/need you identified
2. The material you started with
3. Your design sketch
4. The making process (what worked, what didn't)
5. The final product and its use
6. Environmental impact: What would have happened to this material if not upcycled?

What You Learned: Design thinking, circular economy, upcycling, sustainable production, creative problem-solving, material properties.

IDEA 49 — DESIGN AN ENVIRONMENTAL DATA POSTER

Where: Home or school

Time: 60–90 minutes

Mess level: None to low

Why Science Communication Matters:

The most important scientific finding in the world does nothing if no one knows about it or understands it. Scientists communicate through papers, presentations, and — for public audiences — through accessible, visual communication. A well-designed infographic poster can change minds, influence policy, and move people to action.

Your Task: Choose one environmental issue. Research the data around it. Design a poster (A2 or A1 size — or digital) that communicates the most important facts clearly, accurately, and compellingly.

Possible Topics:

- Plastic pollution in the ocean: scale, sources, impact on marine life
- Deforestation: rates, causes, ecosystem consequences
- Air quality in cities: measurements, health impacts, sources
- Biodiversity loss: species extinction rates, causes, consequences
- Water scarcity: which regions, why, future projections
- Climate change in your country: temperature trends, sea level rise, extreme weather

Design Your Poster:

- **Lead with the most important finding.** What is the one number, trend, or fact that people must know? Make it big and bold.
- **Use visuals that represent scale.** The human brain understands comparisons better than raw numbers. "Enough plastic is produced each year

to fill 100,000 Eiffel Towers" is more comprehensible than "380 million tonnes."

- **Be accurate.** Use data from credible sources: IPCC, UNEP, WHO, Our World in Data, national environmental agencies. Cite your sources.
- **Keep it simple.** Three key messages, clearly communicated, are more powerful than twelve messages that overwhelm the reader.
- **Be honest about uncertainty.** Science involves ranges, confidence levels, and uncertainty. Communicate this honestly rather than overstating certainty.

Share It: Share your poster with your class, your school, or post it online (with your teacher or parents' guidance). Ask for feedback. Did people understand the main message? Did it change how they think about the issue?

What You Learned: Science communication, data visualization, environmental literacy, research skills, design principles, media literacy.

IDEA 50 — DESIGN YOUR OWN STEM PROJECT TO SOLVE A SCHOOL PROBLEM

Where: Wherever your problem is!

Time: This is an ongoing project — weeks or months

Mess level: Depends on what you make!

This is the Final Idea — and the Most Open-Ended.

For Ideas 1–49, we gave you the question. *Idea 50 asks you to find your own.*

Step 1: Observe Your School (or Community)

Spend one week paying attention — really paying attention — to problems and inefficiencies in your school or neighborhood. Carry your science journal and write down everything that seems problematic, wasteful, uncomfortable, unfair, or just harder than it needs to be.

Some possible areas to observe:

- Energy use (lights left on, inefficient cooling/heating)
- Water use (leaky taps, long handwashing queues)
- Waste (where does trash go? Is anything recycled?)
- Food (cafeteria waste? Packaging?)
- Learning (something that's hard to understand because there's no good visual aid?)
- Accessibility (is the school easy to navigate for everyone?)
- Safety (a place that gets slippery when wet? Poor lighting?)

Write down **everything**. Don't judge whether it's "*important enough*." You're observing.

Step 2: Choose Your Problem

Look at your list. Which problem:

- Matters to more than one person?
- Seems like it **could** be solved?
- Connects to something you find interesting?

Choose one problem and write it as a clear **problem statement**: "In our school, [*specific problem happens*] because [*reason*], which means [*consequence for people*]."

Example: "In our school, *students waste a large amount of food at lunch* because *portion sizes are fixed regardless of appetite*, which means *good food goes into landfill and money is wasted*."

Step 3: Research

Find out:

- Has anyone else tried to solve this problem?
- What approaches have worked elsewhere?
- What do experts say about this issue?
- What resources and constraints do you have?

Step 4: Design a Solution

Brainstorm at least five possible solutions. Evaluate each:

- Is it feasible with available resources?
- Does it actually address the root cause of the problem?
- Are there unintended consequences?
- Who would need to be involved?

Choose your best solution and plan it out in detail.

Step 5: Build, Test, Improve

Create a prototype — a first version of your solution. Test it in the real situation. Gather feedback. Improve. Test again.

Step 6: Present Your Work

Create a presentation that includes:

1. The problem statement (with evidence — data you collected)
2. Your research into existing approaches
3. Your solution design
4. How you tested it
5. Results and evaluation
6. What you would do next

Present to your class, your teachers, or your school administration. Real change often starts with exactly this kind of presentation.

You Are Ready.

You have spent this entire book learning to observe, question, experiment, design, build, test, and improve. Those are not just science skills. They are life skills — the skills of someone who sees a problem and says, *I think I can do something about that.*

That is what a scientist is.

That is what an engineer is.

That is what you are becoming.

End of Part 5 — STEM for a Better World

The word "STEM" can feel like an acronym, a category, a career track. But at its deepest, it is something much simpler: it is the human impulse to understand the world and improve it.

That impulse belongs to you.

Final Words

I hope these experiments have challenged you to think, question, and explore like a scientist, engineer, mathematician, or inventor.

Along the way, you may have made mistakes, waited very patiently for results, or watched a trial fail completely. But don't be discouraged — This is a part of how STEM works. The most important thing is you stayed curious enough to keep trying, even though you may not get things right the first time.

You don't need to be naturally gifted at STEM to start. Your skills will grow with practice, and your confidence will grow with experience. Every experiment you complete, every question you ask, and every problem you attempt takes you a step further.

The book may end here, but I hope your exploration does not. Every question you answer leads to new questions waiting to be explored. If there comes a point when you fall in love with a certain subject, definitely go deeper at it. Read. Write. Solve. Trust your curiosity, and allow yourself to explore boldly. No one but you is to decide if something is not for you.

Keep experimenting. Keep exploring. Keep excelling.

The next great discovery could begin with a question you choose to ask.